# Session 2 Overview: *Processors and Communication SoCs*

## DIGITAL ARCHITECTURES AND SYSTEMS SUBCOMMITTEE

**Session Chair:** Shidhartha Das
AMD, Cambridge, United Kingdom

**Session Co-Chair:** Sugako Otani
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Mobile and general-purpose processors continue to push the envelope of performance and power efficiency up to and beyond 5nm. The first three papers show a paradigm shift for mobile computing, followed by generational upgrades in mainstream compute. Latency-critical AI and ray tracing are applied to edge applications in the fourth and fifth papers. The last three papers feature emerging hardware for advanced data communications.

**1:30 PM**

**2.1 A 4nm 3.4GHz Tri-Gear Fully Out-of-Order ARMv9.2 CPU Subsystem-Based 5G Mobile SoC**
Anshul Varma, MediaTek, Austin, TX
In Paper 2.1, MediaTek describes the CPU cluster from its latest 5G smartphone SoC: the first published mobile SoC to feature out-of-order cores in all CPU gears. The 3.4GHz CPUs, built in 4nm, integrate a novel first-level cache design to deliver a 4.6% CPU power reduction, while other power reduction techniques combine to realize a 22% system-level power benefit.

**1:55 PM**

**2.2 "Zen 4c": The AMD 5nm Area-Optimized x86-64 Microprocessor Core**
Srividhya Venkataraman, AMD, Santa Clara, CA
In Paper 2.2, AMD details a same-node area-optimization of its 5nm Zen4 CPU core. Targeting processor density in a 73mm$^2$ die, the optimized core features a 35% smaller L2 cache and 25% smaller CPU core for >25% improvement in performance/mm$^2$ and 9% improvement in performance/W, enabling a 33% increase in cores per package.

**2:20 PM**

**2.3 Emerald Rapids: 5th-Generation Intel® Xeon® Scalable Processors**
Ashley O. Munch, Intel, Hudson, MA
In Paper 2.3, Intel presents Emerald Rapids, the 5$^{th}$-generation Xeon® Scalable Processors built on the Intel 7 process node. Featuring up to 64 cores and over 300MB of shared L3 cache, the multichip package delivers iso-power performance uplift of 18%/24% for integer/floating-point workloads, respectively, as compared to the prior generation.


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2:45 PM

**2.4 ATOMUS: A 5nm 32TFLOPS/128TOPS ML System-on-Chip for Latency Critical Applications**

Chang-Hyo Yu, Rebellions, Seongnam-si, Korea

In Paper 2.4, Rebellions describes their SoC for latency-critical AI inference tasks that require prompt response. Supporting up to 32TFLOPS (FP16) and 128TOPS (INT8), the 5nm SoC delivers a 0.239ms latency for ResNet50 single stream.

3:35 PM

**2.5 A 28nm Physical-Based Ray-Tracing Rendering Processor for Photorealistic Augmented Reality with Inverse Rendering and Background Clustering for Mobile Devices**

Shiyu Guo, Northwestern University, Evanston, IL

In Paper 2.5, Northwestern University presents a 28nm 3.56mm$^2$ physical-based ray-tracing rendering processor for photorealistic augmented reality. Their work enables photo-realistic real-time physical-based ray tracing on mobile devices by embedding end-to-end inverse rendering that achieves 3.95-to-28.8× higher render efficiency than existing ASICs.

4:00 PM

**2.6 A 131mW 6.4Gbps 256×32 Multi-User MIMO OTFS Detector for Next-Gen Communication Systems**

Tang Lee, National Taiwan University, Taipei, Taiwan

In Paper 2.6, National Taiwan University demonstrates a multi-user MIMO OTFS detector, featuring Doppler spread resilience, for next-generation communications systems. The 40nm 9.73mm$^2$ chip delivers a maximum throughput of 6.4Gbps, representing a 3.3-to-18.3 increase over state-of-the-art OFDM detectors, while also realizing a 2.2-to-30.1× improvement in normalized energy.

4:25 PM

**2.7 BayesBB: A 9.6Gbps 1.61ms Configurable All-Message-Passing Baseband-Accelerator for B5G/6G Cell-Free Massive-MIMO in 40nm CMOS**

Wenyue Zhou, Southeast University, Nanjing, China; Purple Mountain Laboratories, Nanjing, China

In Paper 2.7, Southeast University details BayesBB, a 19mm$^2$ configurable all-message-passing baseband accelerator for beyond-5G/6G cell-free massive MIMO in 40nm. Achieving a throughput of 9.6Gbps, a latency of 1.61ms, and integrating 10 Gbps SERDES links, the SoC supports >100bps/Hz spectrum efficiency when tested with a 128×128 MIMO array.

4:50 PM

**2.8 A 21.9ns 15.7Gbps/mm$^2$ (128, 15) BOSS FEC Decoder for 5G/6G URLLC Applications**

Dongyun Kam, Pohang University of Science and Technology, Pohang, Korea

In Paper 2.8, Pohang University of Science and Technology demonstrates the integration of a BOSS decoder as a next-generation FEC for 5G/6G applications. Offering the best solution among recent URLLC FEC designs, this work achieves a latency of 21.9ns and an area efficiency of 15.7Gbps/mm$^2$ in a 28nm technology.

## 2.1 A 4nm 3.4GHz Tri-Gear Fully Out-of-Order ARMv9.2 CPU Subsystem-Based 5G Mobile SoC

Anshul Varma[1], Sumanth Gururajarao[1], HsinChen Chen[1], Tao Chen[1], Gordon Gammie[1], Hugh Mair[2], Jen-Hang Yang[3], Hao-Hsiang Yu[3], Shun-Chieh Chang[3], Cheng-Hao Yang[3], Li-An Huang[3], Kumar Ramanathan[1], Ramesh Halli[4], Efron Ho[1], Ta-Wen Hung[3], Sung S.-Y. Hsueh[3], LiangChe Li[3], Achuta Thippana[1], Ericbill Wang[3], SA Hwang[3]

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Built for upcoming flagship smartphones, this paper describes a new tri-gear CPU subsystem that represents a paradigm shift in mobile SoCs. While tri-gear CPU subsystems have been published [1][2], this work is the first publication to feature Out-of-Order (OoO) CPUs in all three gears of the cluster. Implemented in a 4nm CMOS process, four OoO High Efficiency (HE) cores are complemented by three Balanced Performance (BP) cores, and a single High Performance (HP) core. Critically, full hardware DVFS control is achieved through enhancements to the DVFS control mechanism of prior work [3]. By utilizing an automatic dual-rail equalizer for SRAM supplies and a new High-Bandwidth Voltage Controller (HBVC) for core supplies, all software dependency on power supply sequencing is eliminated. A power benefit is realized through faster response time and by leveraging a power saving mode in the PMIC. Full-CMOS single-rail memories for frequently accessed first-level caches (L1$) deliver significant power savings, while layout techniques for reducing standard cell resistance and capacitance in mid-level layers improve maximum clock frequency. A die photograph is shown in Fig. 2.1.7 and the single-thread power and performance comparison is summarized in Fig. 2.1.1.

In contrast with conventional tri-gear application processor design [1][2], where HE cores are implemented using an in-order micro-architecture (μArch) to minimize logic and standby leakage power, this work utilizes an OoO CPU with a downgraded configuration to improve power efficiency, while achieving the significantly higher Instruction-Per-Clock (IPC) rate of the OoO core μArch. The HE core is implemented with an ARM Cortex-720, configured with a 256KB Level 2 cache (L2$), and clocked at 2GHz using a low-leakage logic library. Thanks to the larger L2$ size and higher IPC derived from the OoO μArch, this HE core achieves performance parity with the previous HE core [2], while clocked at a lower frequency, resulting in up to 67% lower power consumption. The BP and HP cores, previously using Cortex-715 and Cortex-X3 respectively, are both migrated to a Cortex-X4 design. The BP cores, each with a 512KB L2$ and clocked at 2.85GHz, increase performance of multi-threaded workloads by 23% and reduce power by up to 35% at parity per-core performance. Focused on single-thread performance, the HP core with 1MB of L2$ is clocked at 3.4GHz and achieves 16% performance improvement over the prior generation.

For improved power efficiency, the frequently accessed memories of L1$ in HP and BP CPUs are implemented as full-CMOS single-rail arrays featuring a 12T bitcell that eliminates write contention and read disturbance vulnerability (Fig. 2.1.2). With dedicated read and write bitlines, read circuitry is decoupled from the write, enabling independent optimization for performance and power. In traditional 6T SRAMs, the pass-gate transistors are physically shorted on the bitlines increasing the capacitance switched during read and write accesses. In addition, for high frequency clocking, 6T SRAMs traditionally use sense-amplifiers, which further increases switching capacitance during read. Sense-amplifier timing is tuned to guarantee minimum required signal differential with a worst-case bitcell whose read current is derated to cover 6-sigma distribution. However, for a typical bit the same sense-amplifier timing develops a significantly larger differential which further degrades read energy. In contrast to 6T SRAMs, the read circuit in the full-CMOS array hierarchically combines read bitlines. An energy-efficient memory of M*N*P*Q words is implemented by recombining read-word-lines and bitcells through M-input AOI, N-input NAND, P-input NOR, Q-input NAND. Despite an increased number of gate delays, this topology can support HP CPU frequency up to 256 words (M=N=P=Q=4). Scaling to larger word-counts would degrade performance due to increased fan-in, coincidentally the BP/HP μArch partitions L1$ into instances of ≤256 words for other considerations. During read access, only one of the fan-in legs of AOI, NAND, NOR stages can switch resulting in reduced switching capacitance compared to sense-amplifier-aided 6T read circuits. Further, note that the hierarchical read bitlines of the full-CMOS topology only switch if the read/write value opposes the idle value. 6T SRAMs, however, must always discharge one of the complementary bit-line pairs during read or write. As shown in Fig. 2.1.2, there is a 3.7×/73% reduction in transistor fin count switched per output during read access compared to an equivalent 256-word 6T SRAM configured with 64 bits-per-bitline and a 2:1 column mux. Write bitlines are finely segmented at 16b per write bitline, resulting in 2×/50% reduction in switching capacitance during write. There is an overall 2.5×/60% reduction in switching capacitance based on weighted average of read and write accesses for typical workloads. The elimination of a dedicated and/or higher voltage supply makes the full-CMOS arrays DVFS friendly and further enhances energy efficiency at lower CPU voltages. Larger memory instances of the CPU with sparser accesses are implemented with traditional 6T SRAMs utilizing a foundry bitcell. With a power/area trade-off in applying full-CMOS arrays on only the most frequently accessed instances of L1$, CPU total energy consumption is reduced by 4.6% for typical workloads.

When compared to prior technology nodes, 4nm logic cell middle-end-of-line (MEOL) layers resistance and capacitance show a significant increase. This impacts the performance of high-speed applications in the SoC, especially the HP CPU core. To achieve 3.4GHz CPU performance, we explore standard cell design enhancements to overcome this degradation. Fig. 2.1.3 shows some of these standard-cell design features including (a) *split double height* for resistance reduction by replacing a high-R metal0 connection with a shorter metal1 connection, (b) *must join* for back-end via resistance reduction from single-via connection and metal-to-multiple connections of via and metal. Also, we identify frequently used cell pairs based on analysis of top timing-critical paths and design (c) a high-speed combo cell by optimizing the metal0-via0-metal1-via1-metal2 router connection with cell-internal metal0 routing. This results in a higher-performance merged cell (Cell A + B) compared to discrete first and second stage cells.

With dynamic workloads, the energy-aware scheduler (EAS) can update frequency requirements for each CPU in the cluster as often as every 10μs. A hardware-based dynamic frequency & voltage modulation system maintains a CPU group (CPUs on a shared buck form a group) at an optimal operating point for best power/performance. Frequency-locked loop (FLL) with a digitally controlled ring oscillator (ROSC) [3] is used to maintain optimal frequency at a given voltage. While the prior system used software for Adaptive Voltage Scaling (AVS) under changing operating conditions, this work utilizes a novel high-bandwidth voltage controller. Using the target frequency, and the current frequency and ROSC code configuration, HBVC computes the necessary voltage delta to reach an optimal operating voltage for a CPU group and relays this information to PMIC through a dedicated SPI interface. Closed-loop stability tuning mitigates voltage overshoot/undershoot and avoids oscillation. The ROSC code is adjusted continuously based on voltage and temperature for margin reduction. For higher system throughput, latency of the on-die voltmeter is improved over prior work [4], while the addition of a digital on-die sensor network provides temperature information. Fig. 2.1.4 shows silicon measurements of how CPU frequency tracks a changing frequency target with both FLL and HBVC in action. With a 30μs voltage refresh rate, voltage distribution of hardware and software-based DVFS for a popular gaming scenario demonstrates that the CPU achieves a median voltage reduction of 10mV without impacting performance.

As the voltage update rate increases, controlling the SRAM array supply (CVCC) with respect to the peripheral supply (VCC) and bitcell minimum (VSRAM) becomes a challenge. To maintain CVCC ≥ max(VCC, VSRAM), dedicated CVCC LDOs are often used. As VCC is modulated by hardware DVFS based on CPU workload, complex control of LDOs can become necessary. To reduce the resources needed to power SRAMs in different DVFS domains, Dual-Rail Equalizer (DREQ) technology is developed to dynamically switch CVCC power between VCC and VSRAM (typically sourced from a buck) (Fig. 2.1.5). A low-power continuous-time comparator monitors the level of both power rails. With VCC > VSRAM, CVCC is automatically switched to VCC based on the comparator output, and vice versa. Therefore, the CVCC always outputs the higher voltage between VSRAM and VCC. Small hysteresis in the comparator prevents unnecessary toggling between supplies. DREQ eliminates the need for dedicated LDOs and related passive components, which significantly simplifies the power distribution network and reduces the overall BOM cost.

Together, these techniques yield a CPU sub-system that not only adjusts to changing workloads quickly providing power/performance benefits, but also improves resilience under varying operating voltage conditions. Additionally, a power benefit is realized by disabling PMIC active pull-down during a decrease in programmed voltage, allowing the supply to decay as a function of load, as opposed to maintaining a specific slew rate (Fig. 2.1.6). With HBVC, the instances of lower voltages are increased, which contributes to this saving. Fig. 2.1.6 lists power benefit for popular GUI and gaming scenarios. Benefit between 5.98% and 12.25% is realized for a popular gaming application. Finally, as compared to [2]: architectural, circuit, and system innovations of this work combine for a 28% reduction in CPU power and 22% reduction in overall system power for a popular gaming application.

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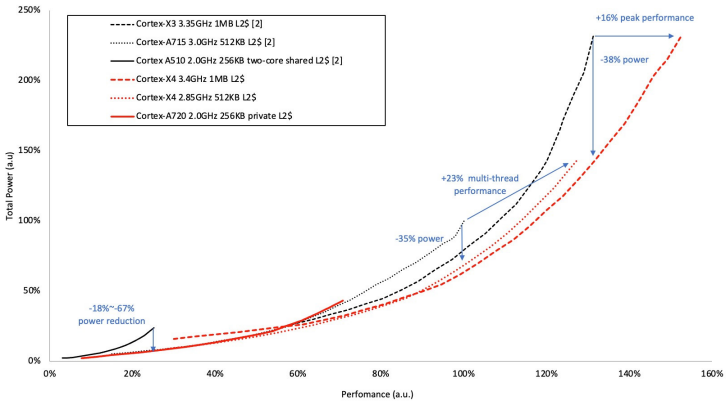


Figure 2.1.1: Single-thread power and performance comparison.

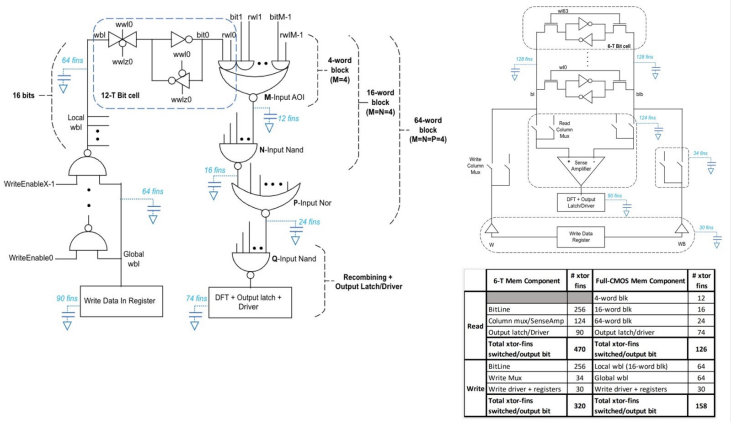


Figure 2.1.2: Full-CMOS array vs conventional 6T memory architecture for L1$.

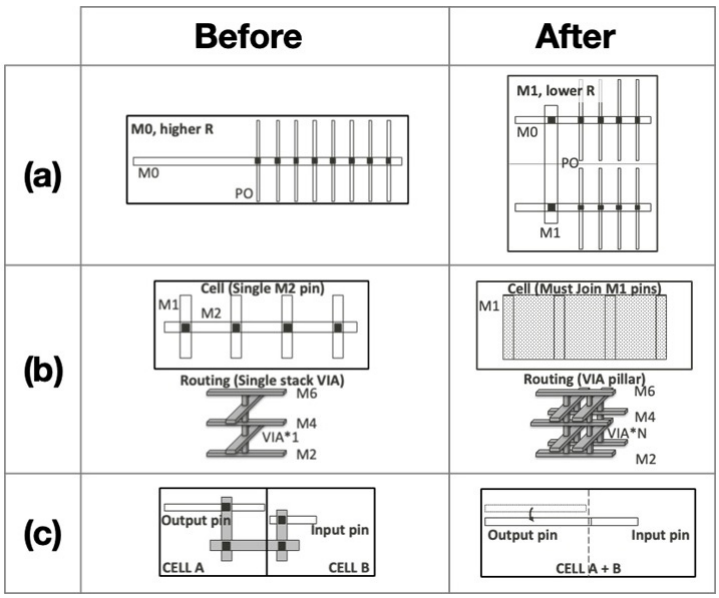


Figure 2.1.3: Standard cell layout enhancements.

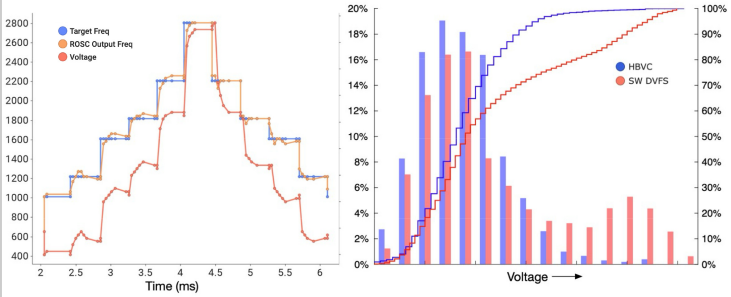


Figure 2.1.4: HBVC and SW DVFS silicon measurements.

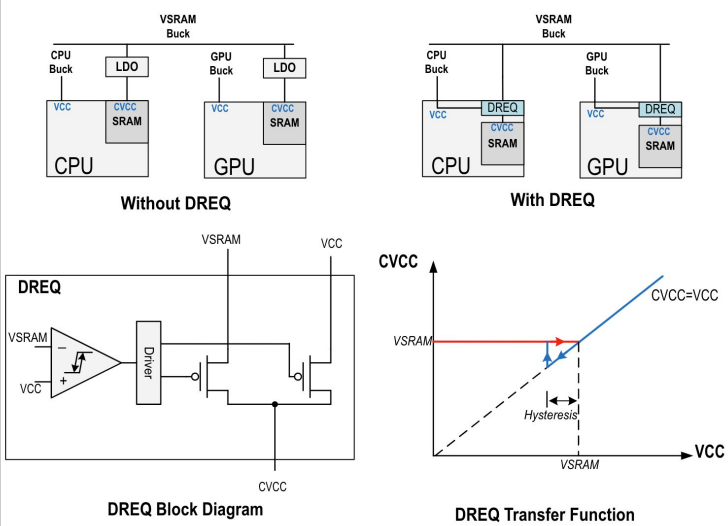


Figure 2.1.5: Dual-rail equalizer.

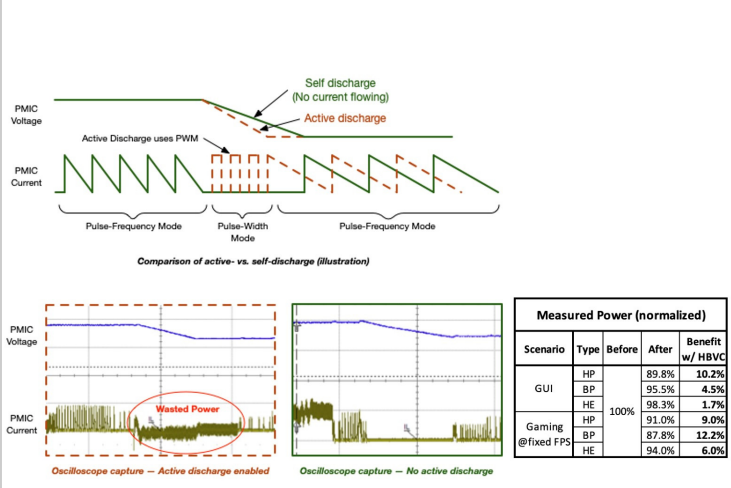


Figure 2.1.6: HW DVFS using HVBC, with and without PMIC active discharge.

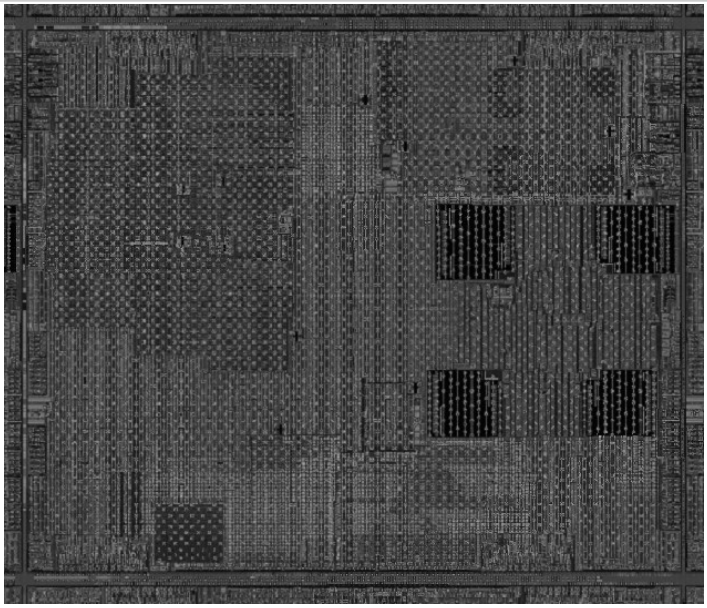

Figure 2.1.7: Die micrograph.


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## 2.2 "Zen 4c": The AMD 5nm Area-Optimized x86-64 Microprocessor Core

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"Zen 4c" is the first area-optimized design in the AMD "Zen" family of microprocessors, targeting energy-efficient and high core-density applications, such as cloud computing. The area of the "Zen 4c" microprocessor core including the private L2 cache is 35% smaller in the same 5nm FinFET process technology [1] as compared to "Zen 4" [2]. With a reduced amount of shared L3 cache per core, from 4MB to 2MB, the core complex die (CCD) of "Zen 4c" has twice the number of cores of the "Zen 4" CCD and the same amount of total L3 cache (32MB), enabling up to 33% more cores per socket. This area optimization enables "Zen 4c" to deliver more than 25% improvement in performance/mm$^2$, and 9% improvement in performance/W on SPECrate®2017_int_base [3], as compared to "Zen 4." "Zen 4c" can operate up to 3.1GHz in frequency in a server configuration, while delivering more power-efficient performance. Fig. 2.2.1 shows the "Zen 4c" based CCD, containing 9.0B transistors in 73mm$^2$ and consisting of two core complexes (CCXs), each of which contains eight "Zen 4c" cores and 16MB of L3 cache.

"Zen 4c" is fully instruction set architecture (ISA) and feature compatible with "Zen 4," both of which have ~14% higher IPC than "Zen 3" [4]. The "Zen 4" and "Zen 4c" architecture provides improved branch prediction accuracy, fetch bandwidth and dispatch bandwidth. A larger Op cache supports improved redirect latency and power with ~50% more entries. The floating-point register file size is also increased and power-efficient support for the 512b Advanced Vector Extension (AVX-512) floating-point instructions is added using a multiplexed 256b data path, providing a power-efficient AVX-512 implementation [4].

Since the first "Zen" microprocessor core, 8T SRAM arrays that can perform one read and one write operation in parallel in a single clock cycle have been heavily utilized in various data structures in the "Zen" family of cores, including the L1 instruction cache, to support the cores' high bandwidth and frequency requirements [5]. In "Zen 4c," the 8T bitcell arrays in the core were largely converted to double-pumped 6T bitcell arrays that are frequently used in graphics engines [6]. The 6T arrays enabled significant area reduction in the core, while the double-pumped architecture maintained the same read/write bandwidth as the 8T arrays. Through aggressive design optimizations, the frequency of the "Zen 4c" double-pumped 6T arrays approached within 20% of the 8T arrays' frequency despite performing the read and write operations serially within a single clock cycle, while providing up to ~40% macro area reduction. To take advantage of the significant macro area reduction, the "Zen 4c" frequency targets were adjusted to accommodate these 6T arrays, and the physical design static timing analysis (STA) frequencies were lowered by roughly 20%. This in turn enabled additional area and power reduction.

This lowering of STA frequencies reduced the total standard cell area by more than 20%, decreasing both the effective switched capacitance (Cac) as well as leakage, and allowed the use of less low-$V_{th}$ transistors to further decrease leakage, while meeting the relaxed STA target. Fig. 2.2.2 shows the shift in the $V_{th}$ mix usage from "Zen 4," which used almost two-thirds low $V_{th}$ devices, to "Zen 4c" in which less than one-third of all transistors were low $V_{th}$. The lower frequency enabled a higher use of multi-bit flipflops, which reduced the input clock pin Cac per flop, and in turn, required fewer/smaller clock gaters to drive the flipflops, which then reduced the Cac loading on the input clock mesh. "Zen 4c" switched from using Design Compiler to Fusion Compiler (EDA tools from Synopsys®), which provided improved capacity to handle larger designs and offered better end-to-end timing correlation. Leveraging this transition, additional area optimization was achieved by using larger tiles as compared to "Zen 4," which in contrast utilized smaller tiles to enable the many more design iterations needed to achieve its higher frequency targets. The smaller core area then shrunk the size of the input clock mesh, creating a virtuous circle of power reduction. The combined optimizations had a significant impact on power, reducing Cac by ~25% and leakage by ~50%.

These area optimizations resulted in a "Zen 4c" core, including the L2 cache, that was 35% smaller than the "Zen 4" core in the same process technology, 2.48mm$^2$ versus 3.84mm$^2$. Side-by-side images shown in Fig. 2.2.3. To fit two CCXs onto one chiplet die, the L3 cache in one CCX was reduced from 32MB to 16MB. The extra frequency margin also allowed pairs of L3 data macros to be combined, amortizing the peripheral circuitry overhead for higher area efficiency, resulting in more than 10% macro area reduction. These two changes enabled a "Zen 4c" chiplet to have twice the number of cores and the same total amount of 32MB L3 cache, with only 10% more area than a "Zen 4" chiplet.

Infinity Fabric™ on the CCD chiplet multiplexes the two CCX L3 interfaces onto a shared Infinity Fabric On-Package (IFOP) link, operating up to 36 Gbps per lane. The 6nm IO die (IOD) chiplet contains the IFOP link transceivers to connect to all the CCDs in package, the external SERDES links, the Infinity Fabric to route data between chiplets and to/from external IO, and other system logic. The IOD is identical between "Zen 4c" "Bergamo" and "Zen 4" "Genoa" server products, demonstrating the incredible value of reusable chiplets to provide highly differentiated products. The IOD chiplet supports up to 12 DDR5 channels at 4800MT/s, and up to 128 PCI Gen5 lanes at 32Gbps per lane. 64 PCI Gen5 lanes can be configured for CXL 1.1+ memory expansion, providing a cost-effective and significant increase in memory bandwidth and capacity. The area-optimized "Bergamo" products are socket and TDP compatible with "Genoa" products, and "Bergamo" can be configured with up to 8 CCD chiplets to deliver up to 128 cores for cloud-optimized server products.

Figure 2.2.4 shows the measured frequency vs. voltage, showing about a 20% reduction in frequency. Fig. 2.2.5 shows the measured frequency vs. power, and the curve for "Zen 4c" has shifted down and to the left, improving the power efficiency at lower frequencies, and extending below the dynamic range of the "Zen 4" frequency-power curve to enable higher performance at ISO power for cloud-native server applications, which typically operate at lower IPC and power than the high-performance applications targeted by "Zen 4."

Figure 2.2.6 shows the measured SPECrate®2017_int_base for the "Zen" family of server products. While "Zen 2" and "Zen 4" had the significant advantage of process technology scaling along with an increase in TDP to deliver large step increases of performance, both "Zen 3" and "Zen 4c" were constrained to the same technology and TDP as their predecessors. While "Zen 3" achieved ~19% SPECrate®2017_int_base uplift with major architectural enhancements and frequency improvement, "Zen 4c" was able to achieve almost half that uplift (~9%) with only an area-optimized physical design and much lower overall design effort. The SPECpower_ssj®2008 benchmark provides a measure of energy efficiency. The 128-core AMD EPYC™ 9754 ("Bergamo") achieved a score of 33,300 operations/s/W, delivering ~9% higher energy efficiency than the 96-core AMD EPYC 9654 ("Genoa"), which achieved a score of 30,602. Both were measured on a two-socket (2P) board configuration.

Figure 2.2.7 shows the normalized competitive performance against commercially available x86_64 (Xeon® 8490H) and ARM® (Altra Max M128-30) server processors on several benchmarks targeting cloud workloads. All measurements were made on 2P systems. Against the x86_64 competitor, the 128-core AMD EPYC 9754 ("Bergamo") processor has up to 2.6× higher performance, and on average has 1.9× more performance across the 8 benchmarks. Against the ARM competitor, the AMD EPYC 9754 has up to 3.7× higher performance and on average has 2.8× more performance. On the SPECpower_ssj®2008 benchmark, the AMD EPYC 9754 is ~1.9× more energy efficient than the x86_64 competitor, and ~2.7× more energy-efficient than the leading ARM competitor.

The area-optimized "Zen 4c" core allows the same 4th generation of the AMD Zen processor family to target fundamentally new server markets by shifting the frequency vs. power curve as shown in Fig. 2.2.5, targeting cloud server and edge applications that have lower frequency requirements than traditional high-performance computing, which requires maximum per-thread performance. In cloud and edge applications, value is provided by maximizing total socket throughput, and the area-optimized "Zen 4c" delivers that with significant improvement in performance/mm$^2$ and performance/Watt.

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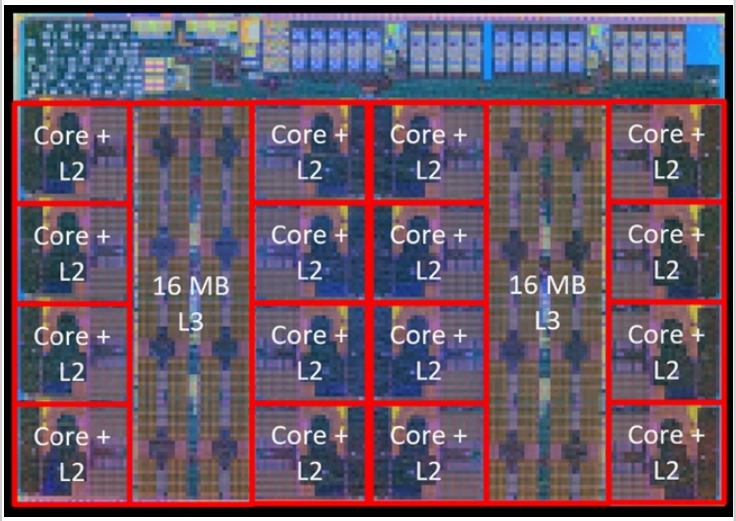


Figure 2.2.1: Die photo of "Zen 4c" CCD.

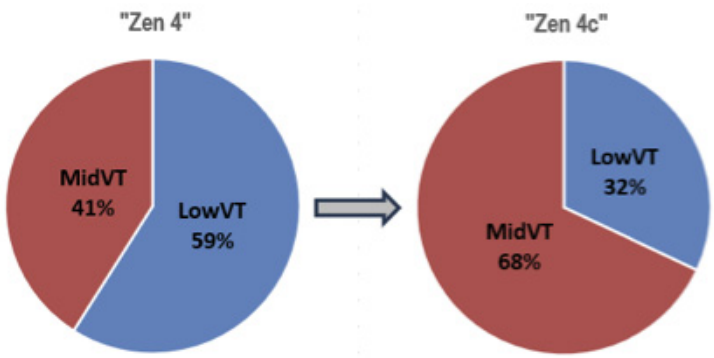


Figure 2.2.2: $V_{th}$ mix breakdown.

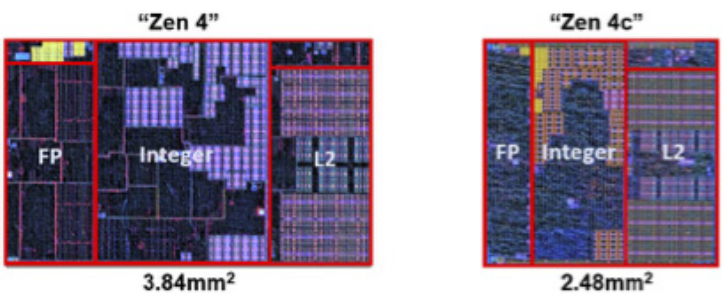


Figure 2.2.3: "Zen 4c" vs. "Zen 4" core + L2 cache floorplan.

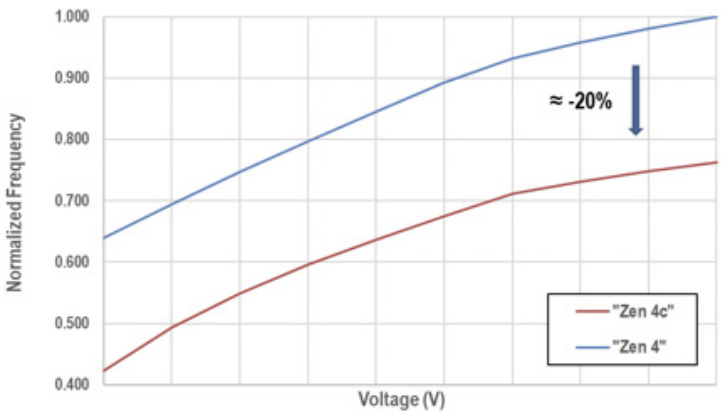


Figure 2.2.4: Frequency vs. voltage, "Zen 4c" vs. "Zen 4".

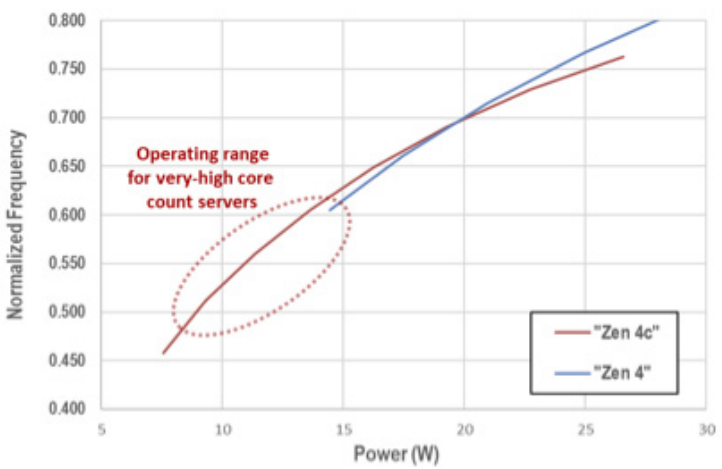


Figure 2.2.5: Frequency vs. power, "Zen 4c" vs. "Zen 4".

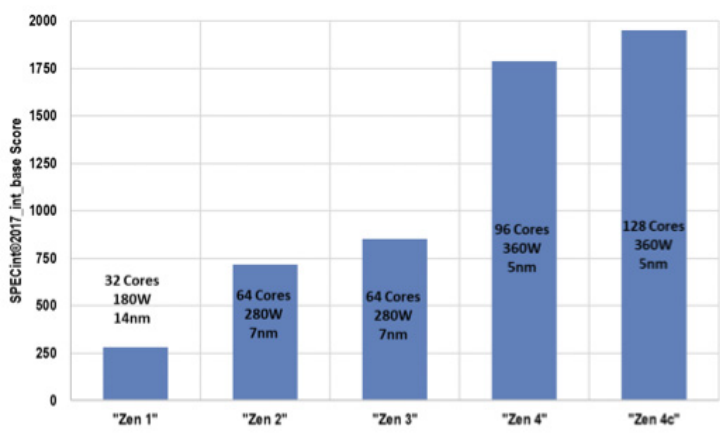


Figure 2.2.6: SpecInt17 comparison.

| | | Normalized Scores | | |
|---|---|---|---|---|
| | | AMD EPYC™ 9754 (Bergamo) | Xeon 8490H | Altra Max M128-30 |
| | Cores | 128 | 60 | 128 |
| *Performance Benchmarks* | NGINX | 2.65 | 1.00 | 0.89 |
| | Redis (get) | 1.49 | 1.00 | 0.56 |
| | Redis (set) | 1.58 | 1.00 | 0.63 |
| | Cassandra-test | 2.04 | 1.00 | 0.68 |
| | Memcached | 1.94 | 1.00 | 0.74 |
| | FFmpeg | 2.27 | 1.00 | 0.79 |
| | MySQL TPC-C | 2.01 | 1.00 | 0.81 |
| | SPECjbb® 2015-MultiJVM | 1.75 | 1.00 | 0.47 |
| | AVERAGE | **1.97** | **1.00** | **0.70** |
| | SPECpower_ssj® 2008 | **1.97** | **1.00** | **0.72** |

Results may vary due to factors including system configurations, software versions and BIOS settings. As of 6/13/2023, see Cloud Native Workloads https://www.amd.com/system/files/documents/amd-epyc-9004-pb-cloud-native-workloads.pdf.

**Figure 2.2.7: Competitive performance and energy efficiency on cloud native applications.**


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## 2.3 Emerald Rapids: 5th-Generation Intel® Xeon® Scalable Processors

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[2]Intel, Fort Collins, CO
[3]Intel, Hillsboro, OR
[4]Intel, Santa Clara, CA

The 5th-Generation Intel Xeon Scalable Processors (codenamed Emerald Rapids) support up to 64 cores, greater than 300MB shared L3 cache, 8 DDR5 channels at 5600MT/s with 1DPC, 32GT/s PCIe/CXL lanes, 20GT/s UPI lanes composed of 2 die (Fig. 2.3.1) in a multichip package. This generation delivers an 18% performance improvement for general integer compute workloads and a 24% improvement for floating-point workloads at iso power vs. the 4th-Gen Xeon processors [1]. This is achieved through an improved core, enhanced process technology, increased core count, significantly larger cache, higher DDR memory speeds, die reductions, and power efficiency improvements at idle conditions. Manufacturing is on the Intel 7 process technology optimized for server usage with improved transistor speed and a focus on leakage and dynamic capacitance reduction enabling +3% frequency/W enhancements over the prior generation process revision. The key power-efficiency improvements at idle were achieved through improvements to the fully integrated voltage regulators (FIVR) [2, 3] to decrease the regulator losses at lower utilization, enhanced active idle detection and power savings, and package C6 power reduction.

The maximum configuration (XCC) is composed of 2 die with one being a physical mirror of the other (referred to as XNC and XSC in Fig. 2.3.7). Each die is built as a 7×7 array of modular integration components that are connected with an on-die scalable coherent fabric (SCF) to provide low-latency, high-bandwidth communication throughout the array. Each modular integration component contains a SCF agent which provides access on and off of the fabric, this includes the core tiles (core plus cache, Fig. 2.3.3), memory controllers (MC), PCIe/CXL agents, UPI agents, and on-die cryptographic, compression and data streaming accelerators. The MC occupies a slot in the first and last columns of the array adjacent to the associated DDR PHY with the remaining slots populated with core tiles. To enable full bandwidth to each I/O agent, two dedicated interleaving rows of SCF are used along the northcap of the die allowing parallel paths to/from the array of cores and memory (Fig. 2.3.2). On-die voltage regulators (FIVRs) are arranged in horizontal strips under each row in nearest proximity to the circuitry they are powering.

The XNC and XSC die are connected by a total of 3 Embedded Multi-Die Interconnect Bridges (EMIBs), which are used to extend the on-die SCF cross die. Each EMIB is embedded into the package substrate consuming 2 package routing layers and blocks any direct package to silicon connectivity. As such, all power delivery to circuitry in the shadow of the EMIB must be done by package surface layer, passthrough the EMIB, or cantilever distribution on the CPU die. Due to reliability constraints in packaging and assembly, the length and aspect ratio of each EMIB is limited requiring multiple bridges on each vertical and horizontal die-crossing edge. These gaps between EMIBs are used for FIVRs providing power to the adjacent circuitry. The result is a quasi-monolithic SoC, consisting of a total of 66-printed cores with one core on each die used for redundancy.

The number of cores and cache were extended beyond those of the prior generation, while reducing the number of die and total silicon area in the package by merging the north and south quadrants into hemispheres, which reduced area, power and latency needed to support die-to-die crossing and allowed optimization of redundant components which was not possible with a 4-identical quadrant design [1]. While the individual die are larger relative to a 4-die configuration, the decreased manufacturing defect density and the elimination of multiple die-to-die crossings results in lower total packaging, assembly, and test (PAT) cost.

The Last Level Cache (LLC) is a unified cache architecture shared across all cores with a total size of over 300MB. It is physically constructed as 5MB per core tile with 20 ways and 4K sets (Fig. 2.3.3). The size represents an increase of over 2.5× compared to the previous generation at a 30% higher density without increasing the pipeline latency at iso-frequency. This was achieved through a combination of larger memory array instances with greater area efficiency, LLC level floorplan optimizations, and improved logic area utilization. The LLC is accompanied by a 3.75MB Local Snoop Filter (LSF) with 15 ways and 4K sets and a 1MB Remote Snoop Filter (RSF) with 4 ways and 4K sets. All the arrays are built with 6T SRAMs cells and include read- and write-assist circuits for improved low-voltage operation. Leakage power is minimized through standard array sleep techniques with less than 2% of the data arrays woken up for an access. To reduce Soft Error Rate (SER) impacts, the LLC data arrays are protected by DECTED ECC, while all other arrays are covered with SECDED ECC. Centralized repair with row, column, and bit coverage is utilized on the LLC data arrays to reduce the area overhead and enhance the repair capability; traditional row and column repairs are used in the remaining arrays. The combination of the read/write-assist circuits, ECC coverage, and advanced repair schemes enable a very low operating voltage for the LLC.

The 5th-Gen Xeon processors support 8 channels of DDR5 memory at 5600 MT/s with 1 DIMM per channel (1DPC). Multiple circuit optimizations including MIM capacitance, package routing, and on-die power delivery improvements along with functional optimizations, such as increased Tx/Rx FIFOs depths and finer gain controls of the strobe receiver, enabled the increase to 5600 MT/s over the prior generation. On-die local voltage regulators provide quiet supplies (VddrIdo) to the Tx and Rx clocking circuits (Fig. 2.3.4) to minimize power-supply noise induced jitter primarily for READ and partially for WRITE operations, while allowing for advanced features to mitigate supply droop and enable low Tx jitter and Rx margin (Fig. 2.3.5). The increase in memory speed, together with increased core count and larger LLC increased overall memory bandwidth by 1.14× per socket and 1.2× inter-socket compared with the prior generation.

To ensure robust and reliable operation across all conditions, the design was optimized to specifically account for the effects exacerbated in large multidie systems. This included package power delivery improvements to minimize variation along with design optimizations to cover static voltage gradients, work-load dependent dynamic droops, and on-die thermal gradients (Fig. 2.3.6). Global clock uncertainty charges were minimized by crediting uniform design topology, and to ensure timing robustness of the global clocks careful consideration was made for metal layer-to-layer variation, accurately extracted capacitance in context of top-side metal, and accounting for across-wafer systematic device variation. Nonuniformly designed local clock timing margins were used to account for silicon to simulation miscorrelation that was observed to vary between gate type, size, and loading. Modular Die Fabric Interconnect (MDFI) clock duty cycle was controlled by intermittent correction at the phase-locked loop (PLL) and guard banded for residual aging. With power and performance optimization a key objective, the clock distribution was designed to balance tight skew requirements through the use of robust power efficient design and circuit techniques. This included optimizing the infrastructure clock distribution across the entire die to minimize the number of global drop-off points and changing from a wide to shorter recombination spine topology for high-speed clocks, while still achieving tight skew that reaped up to 20% savings in the global clock distribution power. Even with the physically larger die, low jitter requirements for the LCPLLs in UPI and PCIe were met with differential clock distribution using top metal and additional specialized clock buffers in the distribution.

*Acknowledgement:*
The authors thank all the teams for their uncompromising dedication in bringing this product to market.

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[3] K. Bharath et al., "Integrated Voltage Regulator Efficiency Improvement using Coaxial Magnetic Composite Core Inductors," *IEEE Elec. Components and Tech. Conf.*, pp. 1286-1292, 2021.

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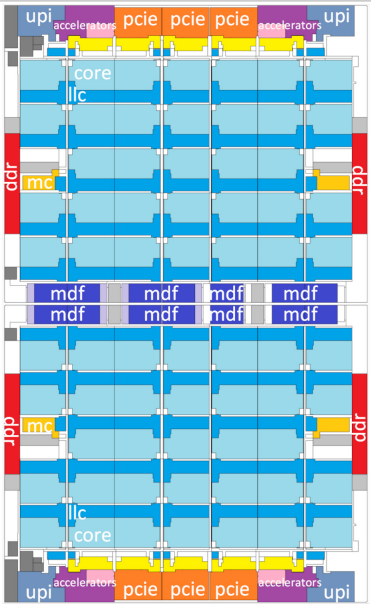


Figure 2.3.1: Processor floorplan diagram for 2-die XCC configuration.

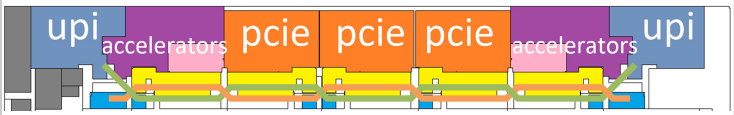


Figure 2.3.2: I/O northcap with 2 interleaving rows of SCF interconnect.

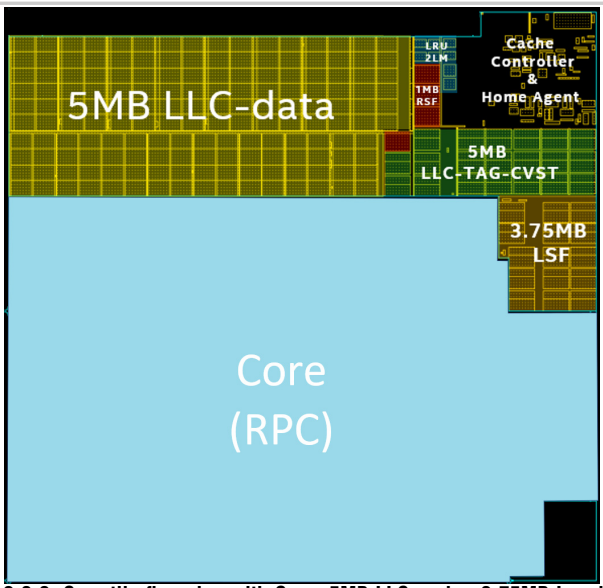


Figure 2.3.3: Core tile floorplan with Core, 5MB LLC cache, 3.75MB Local Snoop Filter (LSF) and a 1MB Remote Snoop Filter (RSF).

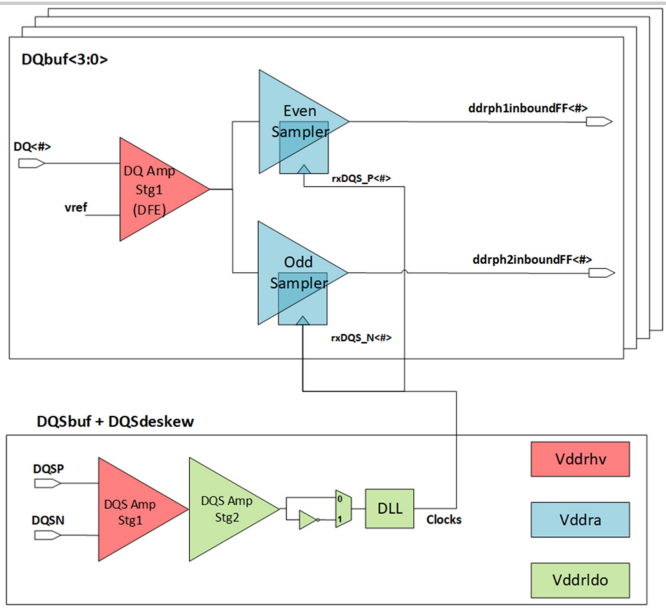


Figure 2.3.4: Voltage domains for DDR receiver data path and clock circuit.

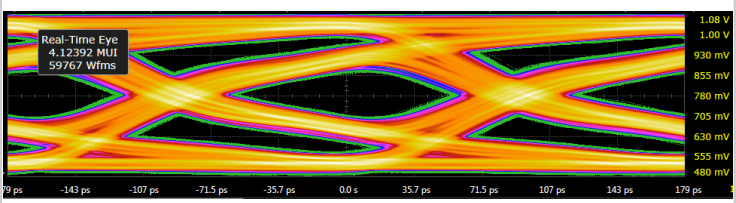


Figure 2.3.5: Silicon measured DDR lane eye diagram at 5600MT/s.

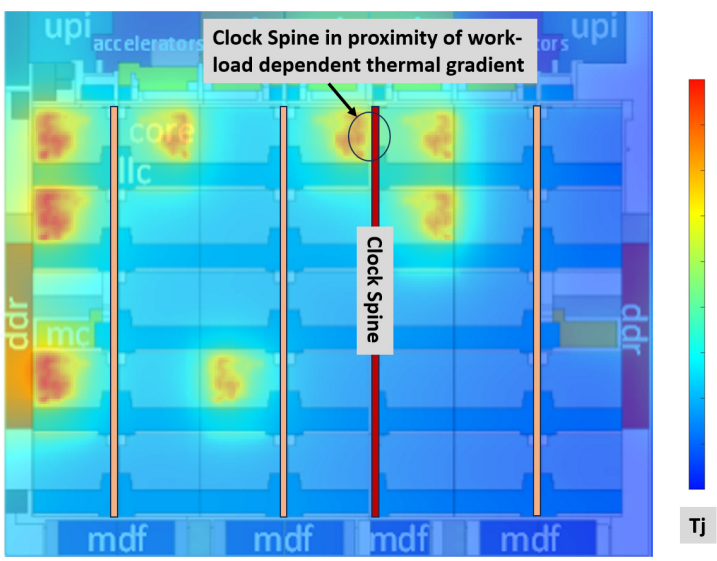


Figure 2.3.6: Illustration of one of the many work-load dependent on-die thermal gradient cases compensated for in design.

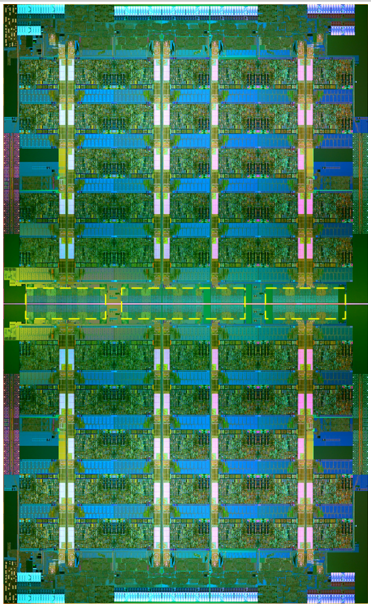

Figure 2.3.7: XNC and XSC die photograph (with EMIB locations highlighted).


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## 2.4 ATOMUS: A 5nm 32TFLOPS/128TOPS ML System-on-Chip for Latency Critical Applications

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Rebellions, Seongnam-si, Korea

The growing computational demands of AI inference have led to widespread use of hardware accelerators for different platforms, spanning from edge to the datacenter/cloud. Certain AI application areas, such as in high-frequency trading (HFT) [1-2], have a hard inference latency deadline for successful execution. We present our new AI accelerator which achieves high inference capability with outstanding single-stream responsiveness for demanding service-layer objective (SLO)-based AI services and pipelined inference applications, including large language models (LLM). Owing to low thermal design power (TDP), the scale-out solution can support multi-stream applications, as well as total cost of ownership (TCO)-centric systems effectively.

Latency criticality in AI-as-a-Service is gaining more attention where batching is not always preferrable, especially for a long pipelined service stack and tight SLO services stacks. We have observed that many cloud services hosted in data centers exhibit the input query distributions from single to multiple streams, reflecting the responsive applications and concurrent input assertion use cases. The different service models in public cloud platforms tend to be more focused on responsiveness and stable 99-percentile latency for industrial-automation solutions.

To realize decreased latency [3] and enhanced resource utilization compared to previous AI accelerators [4-7], the proposed ML inference SoC design comprises flexible AI compute cores and a data/control on-chip network in conjunction with latency-centric memory architectures. Fig. 2.4.1 shows the overall SoC architecture of the AI accelerator. The SoC interacts with a host through a PCIe Gen5 interface with single root IO virtualization (SRIOV of 1 physical function and 16 virtual functions) to provide multiple contexts simultaneously, while ensuring higher resource utilization for multi-tenant users within datacenters and maintaining strict isolation between these contexts. ECC-capable DRAM provides high bandwidth data, and additional on-chip reliability features include logic parity and SRAM SECDED. Root-of-trust (RoT) maintains the SoC security in secure booting, firmware updates and runtime integrity checking. From the AI computing perspective, the SoC consists of two neural engine clusters for ML compute connected with a network-on-chip (NoC) data fabric. Each cluster contains quad neural engines of 16TFLOPS/64TOPS ML compute capability, sharing the L1 ML-centric data cache and cluster interconnect for local data transfer operations, such as all-reduce type operations. As separate cores, the neural engines exploit all instruction, data, and thread-level parallelization across dense compute, memory operations and processor-to-processor communications with decentralized RISC-type processor cores within a single engine. The command processor (CP) compiles command descriptor chains proactively for the given computational graphs. It also oversees task scheduling and dynamic control to maximize the utilization of all SoC components and external resources, while maintaining isolation between concurrent contexts through manipulation of page tables across the entire neural engine. The on-chip storage, organized hierarchically as L0 and L2 memories, holds programs and intermediate data for data sharing, enhancing bandwidth while concealing latency within acceptable limits.

Although many in ML community think that peak FLOPS is a good performance proxy, we believe that keeping the computing units busy is one of the most important factors to build a faster yet efficient inference accelerator. Compared to another AI accelerator [4] or inference-target GPUs [5-7], the SoC achieves a latency improvement of 1.4-3.4× for ResNet-50 and performance efficiency (performance over TOPS) of 1.1-5.5× for Bert-Large, as shown in Fig. 2.4.6. To improve the latency and performance efficiency, multi-layered system features with a balanced architecture are employed. Fig. 2.4.2 shows a multiple level synchronization scheme and a hierarchical memory subsystem architecture. Synchronization is one of the critical factors in terms of latency and utilization due to overheads arising from data dependencies. To address the overhead, five levels of synchronization hierarchy are defined. The L0-sync, the lowest level depicted as s0 in Fig. 2.4.2, resolves the intra-core instruction level and the L1-sync (s1) is targeting inter-core synchronization within a cluster. The L2-sync (s2) tackles synchronization over the cluster and the neural task-level DMA engine managed by the task managers (TMs). The CP handles the coarse granularity task level as depicted in s3, synchronization within the SoC, while dealing with the last level synchronization on an IO coherent interconnect with the host reducing software overhead from coherency, s4. To feed the datapath with required data in a timely manner, a five-level memory subsystem hierarchy has been developed. At the lowest level, depicted as d0 in Fig. 2.4.2, a scratch pad is defined with the highest bandwidth and lowest latency in the SoC. The second level, d1, provides data transfer between nearby engines within a cluster, reducing data bandwidth while preserving latency for streaming and systolic data flow. Fig. 2.4.2 shows a neural cluster cache, d2, which ensures high bandwidth and low latency within the cluster. A shared memory, d3, reduces overall bandwidth requirements in the SoC between the clusters and the external memory, d4. To provide higher effective bandwidth, d3 and d4 are fully interleaved over whole channels. Non-shareable traffic to and from the host fully utilizes the peak bandwidth of the protocol through the non-coherent high-bandwidth interconnect between the host and external DRAM, shown in Fig. 2.4.2 as d4-a. To hit the required memory bandwidth from a single neural engine perspective, we exploit neural data traffic characteristics effectively by the proposed hierarchical memory subsystem as shown in Fig. 2.4.3, d1-d4. The necessary memory bandwidth to the external DRAM can be decreased to 6.25% of the port bandwidth from a single neural engine. This is achieved through the local buses, the L1 neural cache, and the L2 shared memory in which the shared memory is extensively interleaved sub-banked modules.

Apart from the system features, the CP proactively orchestrates the incoming commands from the host to the neural computing and DMA engines to maintain utilization as high as possible. Fig. 2.4.4 shows a typical case for the task manipulation. The CP issues commands that are independent of the data loaded from a host memory and runs them in parallel by utilizing multiple levels of sync mechanisms supported by the SoC, as shown in Fig. 2.4.2. DMA operations can commence whenever the data is ready, as is notified by the two types of sync signals (s4) in task and (s3) in instruction granularity as shown in Fig. 2.4.4. The CP generates tasks and dispatches them after resolving the specific dependencies which need to be managed by the CP. Otherwise, the TM of each neural engine handles inter-task dependencies, (s3), as deep as possible by interacting with other TMs on the SoC. To further optimize the sync latency, a dedicated sync bus is defined between the neural engines, (s1) and (s2), with 16ns for the longest path. A programmable load/store processor, shown in Fig. 2.4.1, interacts with other programmable processors with sync instructions resolving the dependencies, s0. The overall chip utilization across the internal neural engines is improved and the measured performance results are shown in Fig. 2.4.6.

We believe that prioritizing system balance and optimizing the utilization of compute resources, on-chip and off-chip memory, as well as bandwidth, is essential for enhancing inference solutions to support scale-up and scale-out capabilities in datacenter and edge computing. Fig. 2.4.5 shows the comprehensive outputs of the proposed system features. Up to 98.9% neural engine utilization was measured, which is attributed to the command processing control loop with the balanced internal storage and the memory subsystem hierarchy, the scratchpad, the inter-core local bus for streaming data with shareable but less locality, and the neural cache for the data with localities along with the bandwidth amplifying shared memory.

As shown in Fig. 2.4.6, the SoC offers the best latency score for ResNet-50 single stream in MLPerf™ v3.0 for low-profile PCIe cards (under 75W TDP). The chip achieves outstanding performance efficiency for the encoder-decoder transformer network, T5-3B [8], which can be seen as a superset network of decoder-only transformers like GPT-3, LLaMA, and others when compared to other AI accelerator/GPU cards. It is worth noting that this result was obtained using the first version of software (driver and compiler) so a performance improvement can be expected the coming months and years based on other competitors' records (riding the typical software optimization curve in the ASIC domain).

*Acknowledgement:*
The authors wish to convey their profound gratitude for the enthusiastic collaboration and invaluable support received from the entire Rebellions team, all contributors from SEMIFIVE and the exceptional process technology assistance provided by Samsung Foundry.

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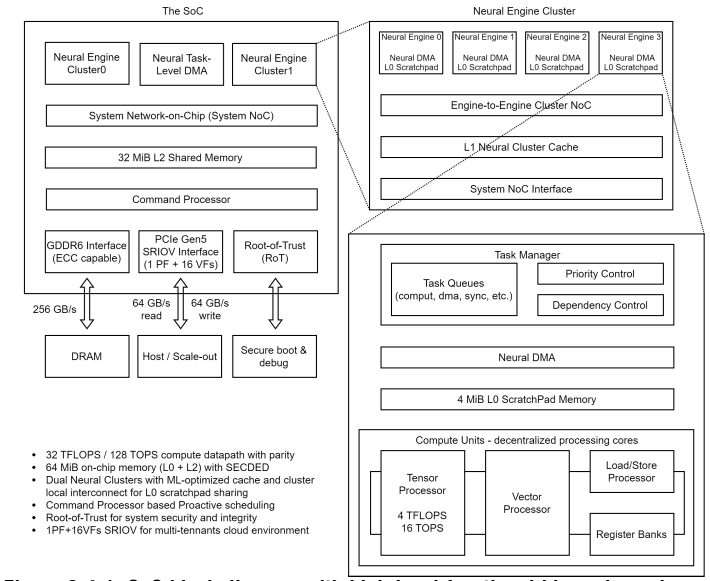



**Figure 2.4.1: SoC block diagram with high-level functional hierarchy and neural engines.**

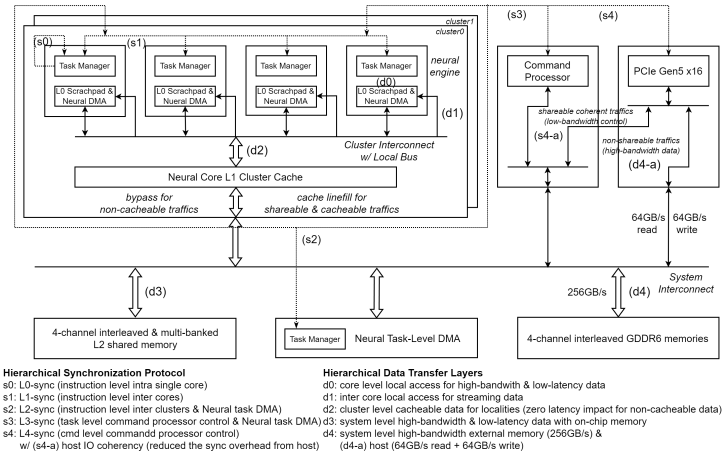
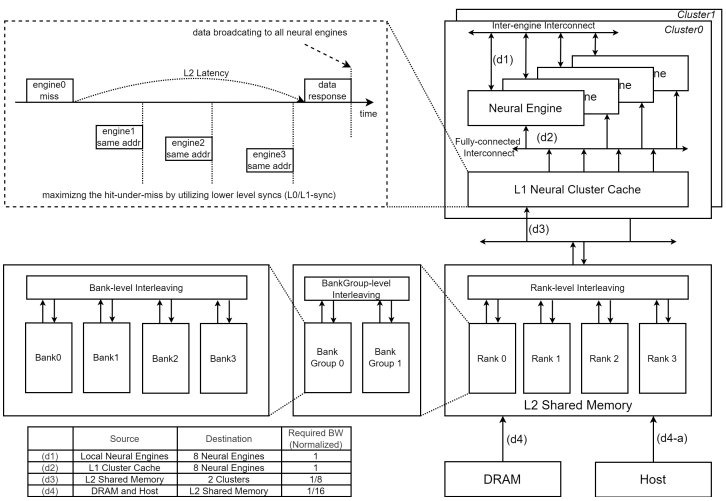



**Figure 2.4.2: SoC system-wide multi-layered synchronizations and the hierarchical memory subsystem.**




| | Source | Destination | Required BW (Normalized) |
|---|---|---|---|
| (d1) | Local Neural Engines | 8 Neural Engines | 1 |
| (d2) | L1 Cluster Cache | 8 Neural Engines | 1 |
| (d3) | L2 Shared Memory | 2 Clusters | 1/8 |
| (d4) | DRAM and Host | L2 Shared Memory | 1/16 |

**Figure 2.4.3: Memory subsystem architecture to meet the required bandwidth in neural engine perspective.**

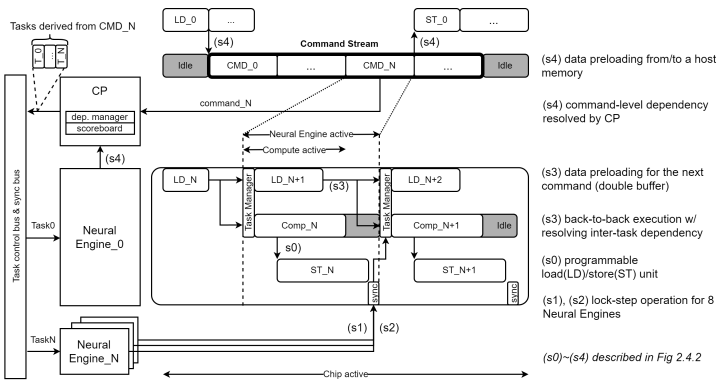



**Figure 2.4.4: Dependency resolving flow with multiple layers of synchronization schemes with CP and TM.**

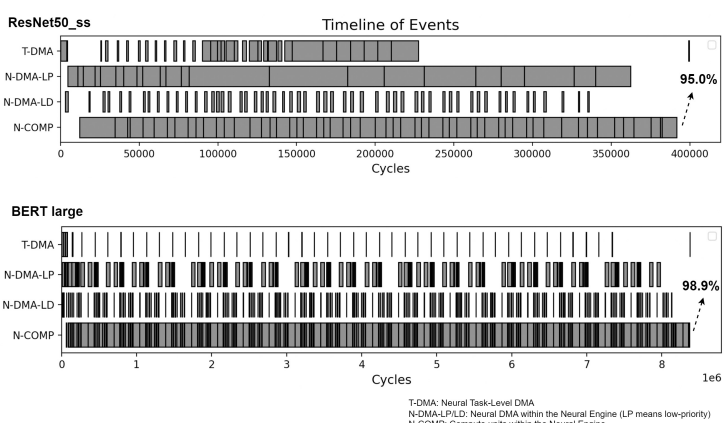
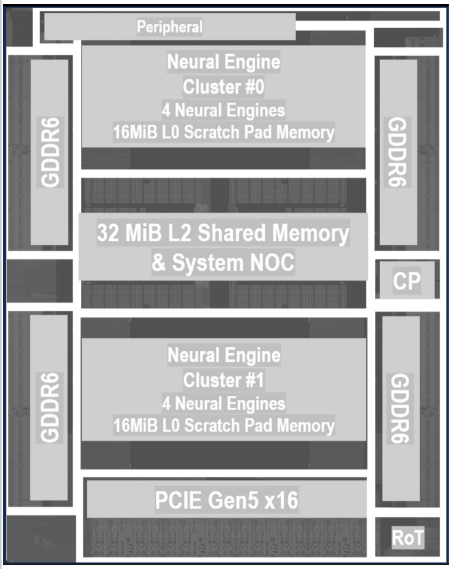


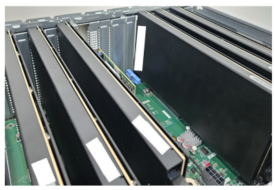

**Figure 2.4.5: Compute Units utilization for MLPerf™ applications, *compute active* described in Fig. 2.4.4.**

| | | NVIDIA T4 [5] ** | NVIDIA L4 [6] ** | NVIDIA A100 [7] ** | Qualcomm Cloud AI100 [4] | This Work |
|---|---|---|---|---|---|---|
| Process | | 12nm | 5nm | 7nm | 7nm | 5nm |
| Area ($mm^2$) | | 545 | 294 | 826 | - | 149 |
| TDP (W) | | 70 | 72 | 300 | 75 | 60 |
| FP16 TFLOPS | | 65 | 121 | 312 | 200 | 32 |
| INT8 TOPS | | 130 | 242 | 624 | 400 | 128 |
| Memory Bandwidth (GB/s) | | 320 | 300 | 1935 | 136.5 | 256 |
| Latency (ms) | Resnet50 ss * | 0.82 | 0.33 | 0.48 | 0.34 | 0.24 |
| | Bert-Large ss * | 6.10 | 2.55 | 1.57 | 7.58 | 4.30 |
| Performance Efficiency (Inferences/sec/TOPS) at single batch *** | Resnet50 ss | 9.4 | 12.5 | 3.3 | 7.4 | 32.7 |
| | Bert-Large ss | 1.26 | 1.62 | 1.02 | 0.33 | 1.82 |
| | T5-3B ENC (512 tokens) | 0.04 | 0.12 | 0.17 | - | 0.43 |
| | T5-3B DEC (1 token) | 1.08 | 0.57 | 0.76 | - | 2.17 |

* Latency results from MLPerf™ [3]
** T5-3B performance is measured at the cloud (GCP) for T4 and L4, and measured at a local server for A100
*** Inferences/sec/TFLOPS is used for T5-3B (FP16).

**Figure 2.4.6: Comparison on the latency and the performance efficiency (inferences/s/TOPS) of ML inference applications.**



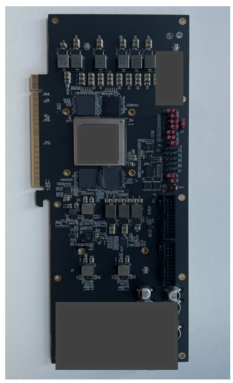


**Figure 2.4.7: Die micrograph and the server platform with the single-slot form-factor PCIe card.**

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## 2.5 A 28nm Physical-Based Ray-Tracing Rendering Processor for Photorealistic Augmented Reality with Inverse Rendering and Background Clustering for Mobile Devices

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As the applications of Augmented Reality (AR) or Virtual Reality (VR) expand rapidly with the growing demands on enhanced visual realism, photorealistic image generation and insertion has become an essential feature for the emerging AR applications providing real-time workplace/household visual assistance. Physical Based Ray-Tracing (PBRT) is often used where synthesized images are generated by simulating the real environment and tracing the light transportation to achieve photorealistic effects, such as reflection, refraction, soft shadows, etc. PBRT is widely used in product design, medical visualization, video games and movie effects. To enable photorealistic rendering, there is a strong demand to support ray-tracing (RT) on mobile devices [1]. However, the challenges are: (1) unstructured memory access pattern and complex control flow lead to scheduling difficulty; (2) high memory requirements exhaust the limited SRAM space on edge devices; (3) low error tolerance requires high precision for computing; (4) complex computations, such as division and square root, require significant computing resources for the edge devices. As a result, common rendering engines such as Apple ARKit, OpenGL, are mainly based on the lower cost rasterization rendering technique. Unfortunately, rasterization rendering fails to produce photorealistic synthesis as shown in Fig. 2.5.1. Few ASICs have been fabricated so far as a mobile photorealistic rendering solution solution, however, they may not support RT [2], or may suffer from low efficiency [3]. This work has developed a ray-tracing processor, which also supports inverse rendering (IR) for background extraction [4]. The key features of this work include: (1) an ASIC rendering processor that embeds an end-to-end PBRT solution with IR for AR on mobile devices, (2) a reconfigurable mixed-precision PE design supporting diverse computing tasks for both IR and RT, (3) background clustered Field of View (FOV)-focused 3D construction reducing conventional background scene complexity from O(nlogn) to O(1), (4) scalable partitioning scheme for complex 3D objects, with an average of 13× speed up on test scenes, (5) use of Global RT Scheduler (GRTS) and Global Memory Access Controller (GMAC) to overcome the challenges of irregular memory access pattern and varied PE run-time with overall 684× speedup compared with the baseline design. The 28nm test chip achieves 3.95-28.8× higher rendering efficiency compared with existing ASIC solutions, enabling real-time PBRT rendering on mobile edge devices.

Figure 2.5.2 illustrates the computing flow of this work. In the first step, a 2D image captured by a regular camera is sent through a CNN-based physical decoder and encoder for IR to obtain the background physical attributes, including four major background physical attribute (PA) maps: albedo, normal, lighting and depth maps. To save the compressed PA map on chip, a background clustering scheme is developed based on the similarity of neighbor pixel values of the background map by applying an average filter. The result PA maps are stored in a Physical Attributes MEM (PAMEM). Each PE accessing the PAMEM passes through the Per Pixel Compression Decoder (PPCD) and the Unified Address Converter (UAC) to fetch the corresponding background physical attributes parameter based on the PE task ID from the GRTS scheme. In this way, 145-4800× of memory saving for different backgrounds is achieved with 0.06% hardware overhead compared with the baseline design. In the second step, the IR result is used for Camera FOV-focused 3D construction. As shown in Fig. 2.5.2, the background scene is constructed only for the 3D space covered by the user camera FOV. In this way, the background scene complexity is reduced from O(nlogn) to O(1) compared with conventional RT solutions (n refers to the number of geometric primitives in the scene) [5]. In the last step, PBRT rendering is implemented to render 3D virtual objects with an average of 76% RT workload reduction compared with the conventional RT solutions as shown in Fig. 2.5.2.

Figure 2.5.3 shows the top-level architecture of this design. To increase the utilization of the PE and address the irregular access pattern to memory, an 8×6 PE array is implemented with a GRTS and a GMAC. To support computation for both IR and RT modes, a reconfigurable mixed-precision PE is developed as shown in Fig. 2.5.3. Each PE contains a local PE Controller, clock-gating control, a computing core which supports 8b, 16b and 32b MAC, 64b division and 64b square root operations and a local OBJMEM. Clock gating disables excessive computing units and local MEM in IR and RT modes with 32% power saving. In RT mode, a scalable 3D model partitioning flow shown in Fig. 2.5.3 is implemented. In contrast to the conventional solution that builds the BBOX acceleration [6], this work introduces two types of object Bounding Box (BBOX): Empty BBOX (EBBOX) and Target BBOX (TBBOX). EBBOX is only used for light transportation estimation and shadow purposes, while the shading computation is skipped. TBBOX includes a sub-group of user-defined objects inside. After the BBOX Intersection Evaluator (BBIE) detects intersection with TBBOX, the Triangle Mesh Intersection Evaluator (TIE) computes the triangle intersection, and the result is sent for shading computing. With this scheme, complex 3D objects could be segmented for RT processing without losing the ray-tracing effect. As a result, a linear scalability in RT rendering time and an average of 13× speed up with only 5.6% memory overhead is achieved compared with the baseline design.

Figure 2.5.4 shows on-chip data movement in IR and RT modes. In IR inference mode, double input and weight stationary are supported. In RT mode, multiple PEs have memory access conflicts, as shown in Fig. 2.5.4. To address the global PAMEM access conflict and varied PE run-time, GRTS and GMAC are implemented. With the RT Token Checker (RTTC) checking one PE status every clock cycle, GRTS and GMAC process the RTTC selected PE request individually while GRTS refreshes the checked PE status if the computation is done. In this way, we achieve 42.8× overall speed up from GRTS and 16× overall speed up from GMAC compared with the baseline design by introducing only 2.8% and 0.6% hardware cost. The detailed RT shading algorithm to compute the color of each pixel is also shown in Fig. 2.5.4. RT shading demands complex operations, such as sqrt and division. In this work, a PE Compute Unit (PCU) is implemented inside each PE. As shown in Fig. 2.5.4, PAMEM, OBJMEM data are sent to PCU for shading computation. PCU's output is stored in a local shading register for lighting effect accumulation. By implementing a PCU in each PE, all the RT computation can be finished individually inside each PE.

Figure 2.5.5 shows the total runtime breakdown for the IR-RT flow and demonstrates the background clustering scheme by showing the background reflection is able to light up the object properly by using clustered PA map compared with the baseline design with the detailed background PA map. Four virtual object insertion test cases with teacup, Utah teapot, Stanford bunny and four spheres are demonstrated in Fig. 2.5.5 using the IR-RT rendering scheme. Different materials, such as glass, mirror and ivory are displayed with photorealistic rendering effects of reflection, refraction and shadow. The IR-RT flow achieves an average of 26fps for real-time RT rendering with complex 3D objects. Figure 2.5.5 also showed the 3D rendering case without IR with a predefined 3D background. Four spheres with different materials are inserted. Photorealistic effects of refraction, reflection and object shadow are properly rendered to the image with 78fps, meeting the requirement of real-time AR applications.

A 28nm test chip was fabricated with 0.9V nominal supply voltage. Figure 2.5.6 shows more measurement and comparison results. Power and frequency scaling are shown in Fig. 2.5.6 with a supply voltage from 0.6V to 0.9V. 500fps/W and 1418fps/W power efficiency has been achieved at 0.9V for IR and RT modes, respectively. The comparison table with prior art is provided in Fig. 2.5.6. This work implements IR-RT-based rendering solutions, achieving 28.8× and 3.95× higher ray-tracing rendering efficiency compared with prior ASIC designs, enabling real-time PBRT on mobile edge devices. Figure 2.5.7 shows the die photo and chip specifications.

*Acknowledgement:*
This work is supported in part by AFRL under the DARPA RTML program under award FA8650-20-2-7009 and NSF grant CCF-2008906.

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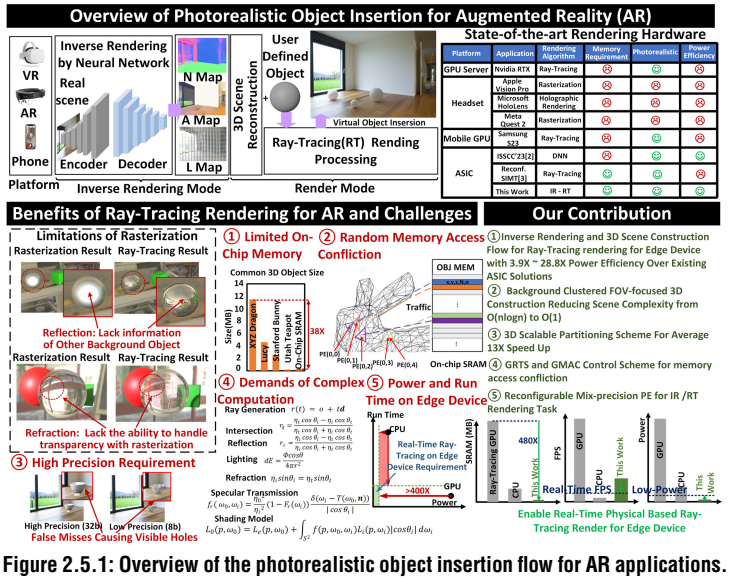



Figure 2.5.1: Overview of the photorealistic object insertion flow for AR applications. Limitation of rasterization. Challenges for ray-tracing rendering and contributions of this work.

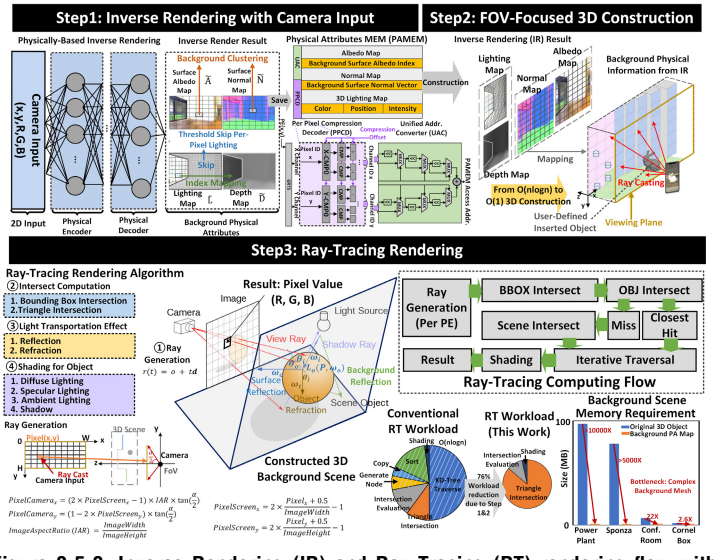



Figure 2.5.2: Inverse Rendering (IR) and Ray Tracing (RT) rendering flow with background physical attributes and camera field of view (FOV)-focused 3D construction.

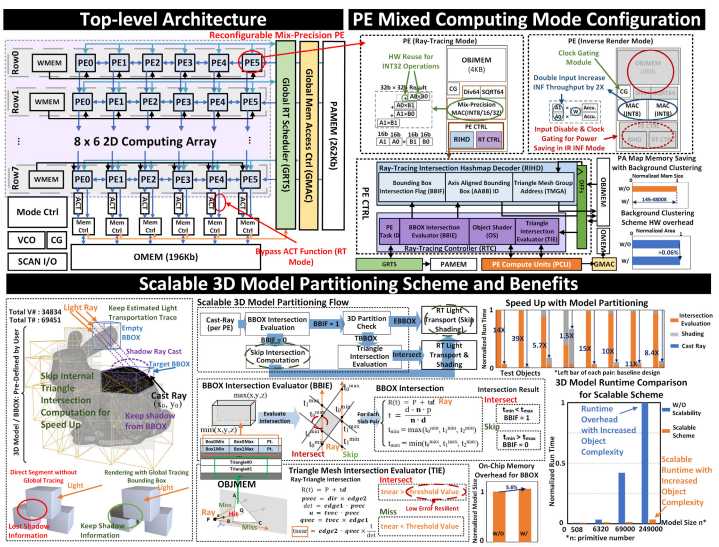



Figure 2.5.3: Top-level chip architecture of this design. Reconfigurable mixed-precision PE architecture and scalable 3D model partitioning scheme. BBOX intersection evaluator and triangle mesh intersection evaluator.

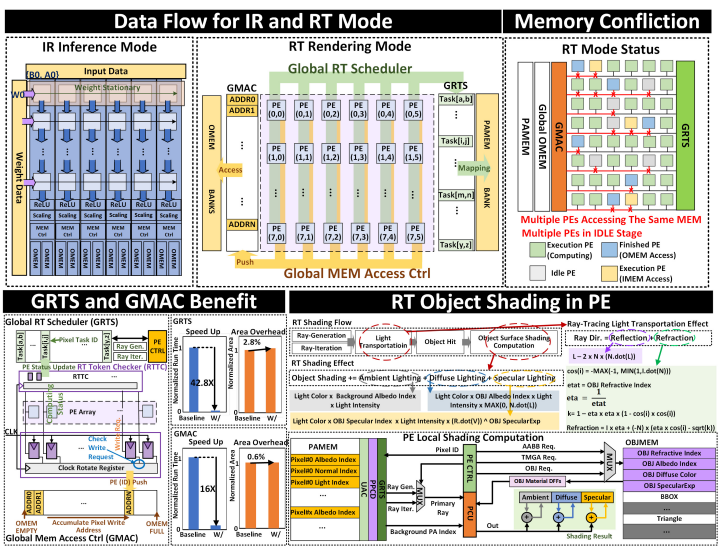



Figure 2.5.4: Global Ray-Tracing Scheduler (GRTS) and Global Mem Access Controller (GMAC) scheme and benefits. Ray tracing object shading computation in each PE.

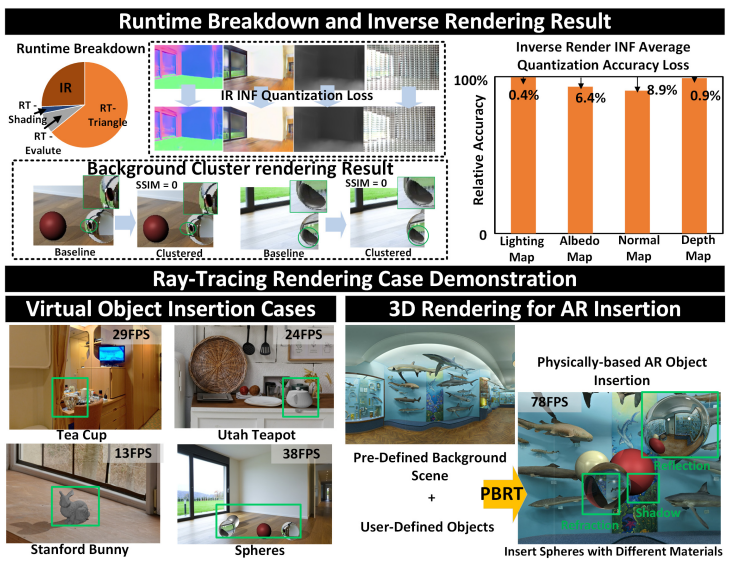
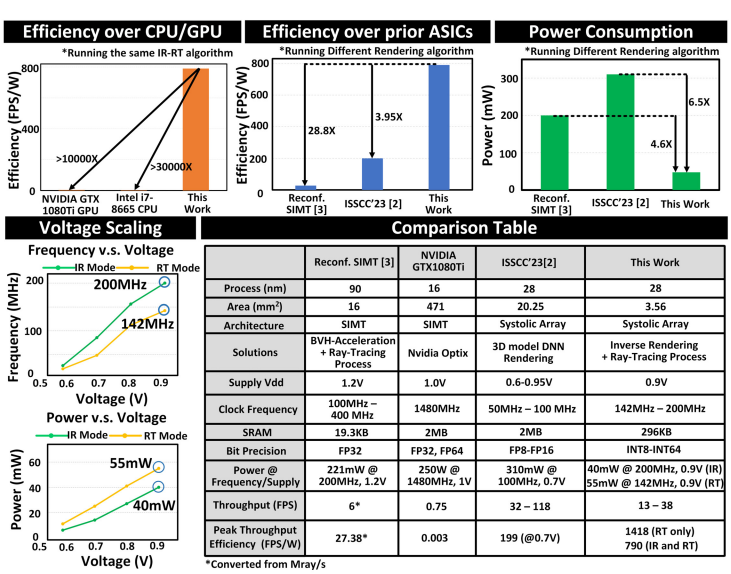


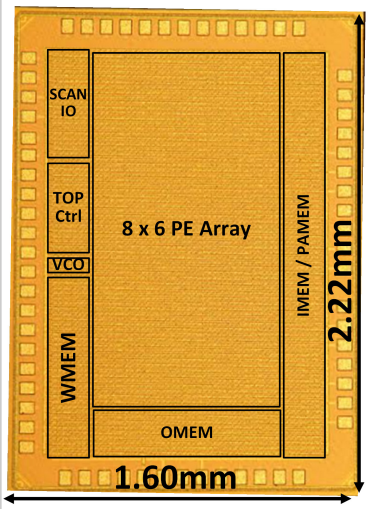

Figure 2.5.5: Inverse rendering quantization accuracy loss. Examples of virtual object insertion with IR-RT flow and 3D rendering for AR object insertion.




| | Reconf. SIMT [3] | NVIDIA GTX1080Ti | ISSCC'23[2] | This Work |
|---|---|---|---|---|
| Process (nm) | 90 | 16 | 28 | 28 |
| Area (mm²) | 16 | 471 | 20.25 | 3.56 |
| Architecture | SIMT | SIMT | Systolic Array | Systolic Array |
| Solutions | BVH-Acceleration + Ray-Tracing Process | Nvidia Optix | 3D model DNN Rendering | Inverse Rendering + Ray-Tracing Process |
| Supply Vdd | 1.2V | 1.0V | 0.6-0.95V | 0.9V |
| Clock Frequency | 100MHz – 400 MHz | 1480MHz | 50MHz – 100 MHz | 142MHz – 200MHz |
| SRAM | 19.3KB | 2MB | 2MB | 296KB |
| Bit Precision | FP32 | FP32, FP64 | FP8-FP16 | INT8-INT64 |
| Power @ Frequency/Supply | 221mW @ 200MHz, 1.2V | 250W @ 1480MHz, 1V | 310mW @ 100MHz, 0.7V | 40mW @ 200MHz, 0.9V (IR) 55mW @ 142MHz, 0.9V (RT) |
| Throughput (FPS) | 6* | 0.75 | 32 – 118 | 13 – 38 |
| Peak Throughput Efficiency (FPS/W) | 27.38* | 0.003 | 199 (@0.7V) | 1418 (RT only) 790 (IR and RT) |

*Converted from Mray/s

Figure 2.5.6: Measurement results and comparison table with prior work.




| | |
|---|---|
| Process | 28nm CMOS |
| Area ($mm^2$) | 3.56$mm^2$ |
| SRAM | 296KB |
| Supply Vdd | 0.6~0.9V |
| Bit Precision | INT8,16,32,64 |
| Max IR Frequency | 200MHz |
| Max RT Frequency | 148MHz |
| IR Power | 40mW @ 0.9V |
| RT Power | 55mW @ 0.9V |
| IR Efficiency (FPS/W) | 500 |
| RT Efficiency (FPS/W) | 1418 |

**Figure 2.5.7: Die micrograph.**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## 2.6 A 131mW 6.4Gbps 256×32 Multi-User MIMO OTFS Detector for Next-Gen Communication Systems

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National Taiwan University, Taipei, Taiwan

Future wireless systems are expected to accommodate a variety of devices with high-mobility [1], such as high-speed rail, unmanned aerial vehicle (UAV), and low-earth-orbit (LEO) satellite, as shown in Fig. 2.6.1. Under such fast time-varying channel conditions, however, the performance of current orthogonal frequency-division multiplexing (OFDM) systems degrades due to inter-carrier interference caused by severe Doppler spread. The orthogonal time-frequency space (OTFS) system has been recognized as a promising solution to high-mobility communication [1]. Unlike OFDM that multiplexes data symbols in the time-frequency (TF) domain, OTFS multiplexes data symbols in the delay-Doppler (DD) domain. In the OTFS system, the transmitted data are modulated through an inverse symplectic finite Fourier transform (ISFFT) and Heisenberg transform. The received data are demodulated through symplectic finite Fourier transform (SFFT) and Wigner transform [1]. The equivalent DD domain channel indicates the physical geometry of the wireless environment, enabling high resilience against Doppler spread. Integration of OTFS technology and massive multi-user multi-input multi-output (MU-MIMO) is envisioned to meet the growing demand for future high-throughput, high-mobility communication systems.

As illustrated in Fig. 2.6.1, the bit error rate (BER) performance indicates that the OTFS system is not subject to Doppler spread and thus is more robust than OFDM in high-mobility scenarios for MU-MIMO cases. However, the superior performance of the OTFS system demands excessively high detection complexity. According to our experiment on a 256×32 MU-MIMO system with 256-QAM modulation, the detection complexity of the OTFS system is up to 32× higher than its OFDM counterpart. To the best of our knowledge, efficient hardware architecture has never been explored in the open literature. This work presents a high-throughput massive MU-MIMO detector in silicon for OTFS systems.

Figure 2.6.2 illustrates the message-passing (MP)-based OTFS detection algorithm [2] for massive MU-MIMO systems, in which the interference is approximated as a Gaussian random variable. Symbols can be estimated by identifying the closest symbol within a constellation set. Based on the Gram matrix of the channel matrix ($H^HH$), a factor graph can be constructed with observation nodes and variable nodes. The interference statistics and symbol estimates are iteratively updated and exchanged through message passing. Soft information can be evaluated by the log-likelihood ratio (LLR) [3] to further enhance the BER performance in a channel-coded OTFS system. According to our experiment, a 2.5dB gain can be obtained by employing soft information for a length-512 rate-1/2 polar code with belief propagation decoding.

In this work, the complexity of the MP detection algorithm is minimized by applying algorithm-architecture co-optimization. The variance is estimated by a scaled maximal squared magnitude of channel coefficients adaptively [3]. The structure of the Gram matrix in the OTFS system is fully utilized to reduce the complexity in estimating variances by leveraging the Hermitian symmetry, the twisted block circulant property of submatrix, and the circulant structure of each block. This reduces the overall MP complexity by 75% while the accuracy of the variances is still maintained. The block sparsity in each submatrix is utilized to reduce the complexity in estimating interference means by skipping the associated computations [2]. Block-wise circular convolution in the DD domain is implemented efficiently by elementwise multiplications in the delay-time (DT) domain by a Fourier transform. This reduces the overall MP complexity by 70% without accuracy loss in estimating means. Layered scheduling is usually applied to improve the convergence speed through layer-by-layer detection, at the cost of additional memory storage. To reduce memory overhead, the residual noise (RN) can be updated recursively by the difference between symbols in the current and last iterations. The memory size is reduced by 80% for memory storage with 4b symbols and 20b partial interferences. Correlated symbols can also be reused for four consecutive layers, saving 72% memory size for the required error performance. The proposed update scheme effectively achieves 2× better convergence speed with minimized memory overhead (6% of the direct-mapped design).

Figure 2.6.3 shows the system architecture of the proposed massive MU-MIMO OTFS detector, which includes a 16-core processing array, an LLR computation unit, and a channel memory bank. Each processing core has an interference cancellation unit (ICU), a mean computation unit (MCU), a variance computation unit (VCU), and a core memory bank. The ICU is used to cancel the interferences in the RN and to generate the symbol estimates. The MCU and VCU respectively compute the interference mean and variance. The core memory bank stores the RN and the reused symbols. The LLR computation unit computes the soft information by piece-wise linear functions. The channel memory bank includes one channel-sample memory and one channel-difference memory for storing the encoded Gram matrix coefficients.

Figure 2.6.4 shows the detailed architecture and optimization techniques for the MCU. The mean values are computed along the diagonal band of the submatrix in a block-by-block manner. It is noted that the mean values for the lower-left and upper-right blocks can be derived from the symbols updated in the preceding layer and pre-computed as partial sums, respectively, to eliminate data dependency. Compared with the row-wise baseline design, the proposed architecture achieves full hardware utilization. The MCU contains 4 complex-valued multiply-accumulate (CMAC) computation units, each including two sum units and two accumulation units for temporal and sequential operations, respectively. FFT and IFFT modules are included to efficiently perform circular convolutions in the DD domain. Moreover, the diagonal-like structure of both the lower-left and upper-right blocks is well handled by hard-wired connection. Since the symbols are estimated more accurately as the MP detection progresses, the maximal input symbol difference of the MCU decreases. This leads to an underutilized MCU with an increased latency. A dual-mode multiplier in the CMAC computation unit is proposed to increase the hardware efficiency. It performs standard multiplications for the earlier iterations and parallel multiplications for the latter iterations. The proposed design reduces the latency for mean computation by 24% in the high-SNR regime (with BER < $10^{-4}$).

Figure 2.6.5 shows the optimized hardware architecture of the VCU, which estimates the interference variance. In the VCU, the DT-domain channel coefficients are converted to DD-domain coefficients whose squared magnitudes are computed to update the variance estimates. Variance computation is conducted over two iterations and thus the VCU can be shared to minimize the area cost. In this work, a layer-ordering scheme, based on layered scheduling, is proposed to reduce the memory access. Layer ordering is efficiently implemented by a round-robin ring interconnect between the processing cores. Unlike the baseline design, which accesses two submatrices, the proposed scheme only accesses one submatrix, reducing 50% of the memory accesses for one processing core. The Hermitian symmetry of the Gram matrix (1.3MB) can be utilized to reduce the channel memory size. The high correlation along the block diagonal direction (due to the twisted block circulant structure) allows approximation of the channel coefficients, which can be encoded by fewer channel samples and differences. In this work, four samples are selected for the approximation. The proposed channel coefficient encoding cuts the memory size and accesses by 89% and 82%, respectively, for the channel memory bank.

The massive MU-MIMO OTFS detector supports MU-MIMO system with up to 32 users, 256 receive antennas, and 256-QAM modulation. Fabricated in 40nm CMOS technology, the chip integrates 6.76M gates in core area of 6.47mm$^2$, including 167KB on-chip SRAM. This work delivers a maximal throughput of 6.4Gbps and the power dissipation of the chip is 131mW operated at a 200MHz operating frequency from a 0.9V supply. Figure 2.6.6 shows the BER performance of the chip and the performance comparison with the state-of-the-art OFDM detectors. As can be seen, the error performance loss of the chip is negligible (0.12dB) when compared to the floating-point, unoptimized implementation [2]. The performance of the OTFS detector is compared with state-of-the-art OFDM detectors [4-6]. This work achieves 3.3-to-18.3× higher maximal throughput and 2.2-to-30.1× lower normalized energy, in addition to the resiliency against the Doppler spread, providing a promising solution for next-generation communication systems with high-mobility. Figure 2.6.7 shows the chip micrograph and summary.

*Acknowledgement:*
This work is supported by NSTC of Taiwan and Intelligent & Sustainable Medical Electronics Research Fund in NTU. The authors also thank TSRI for technical support on chip design and fabrication.

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High-Speed Rail

Unmanned Aerial Vehicle

Low-Earth-Orbit Satellite

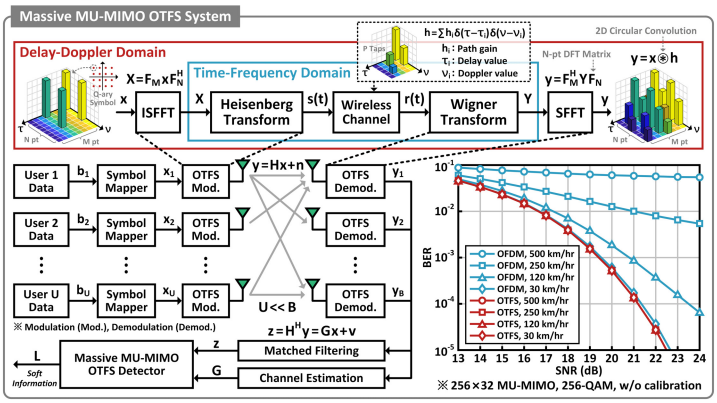



**Figure 2.6.1: Massive multi-user MIMO OTFS system for high-mobility communications.**

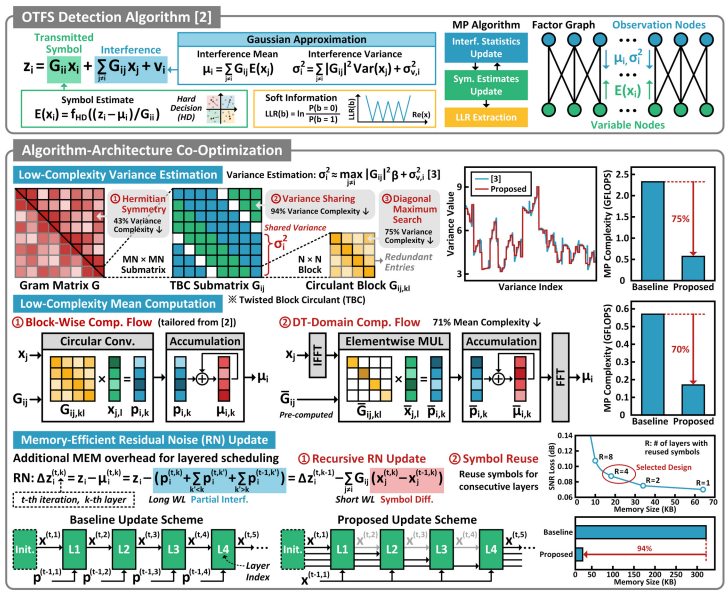
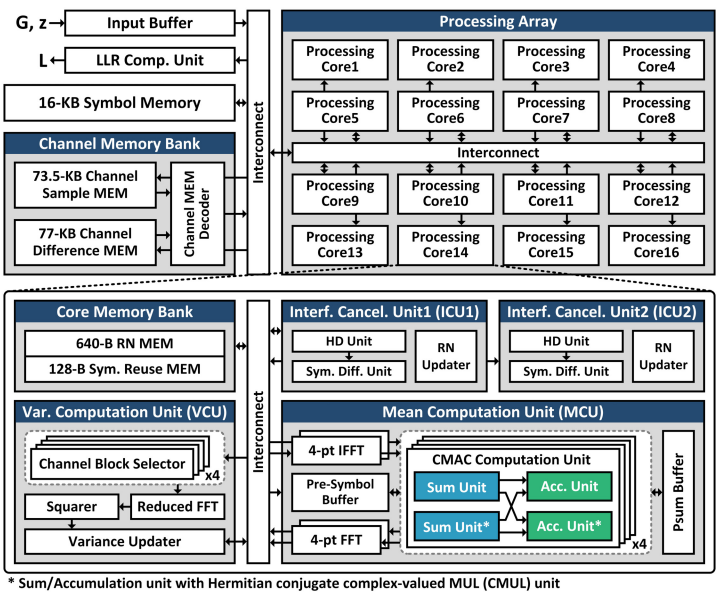



**Figure 2.6.2: OTFS detection algorithm and algorithm-architecture co-optimization.**




* Sum/Accumulation unit with Hermitian conjugate complex-valued MUL (CMUL) unit

**Figure 2.6.3: System architecture of the proposed massive MU-MIMO OTFS detector.**

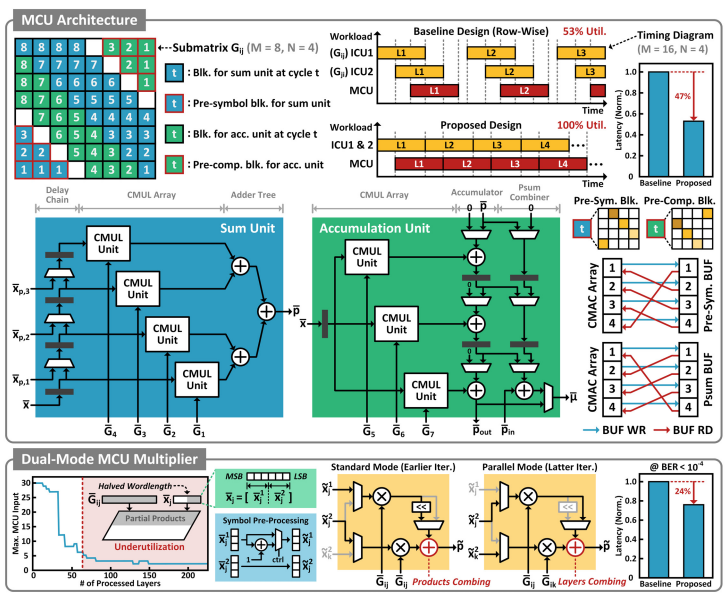



**Figure 2.6.4: Proposed MCU architecture and dual-mode MCU multiplier.**

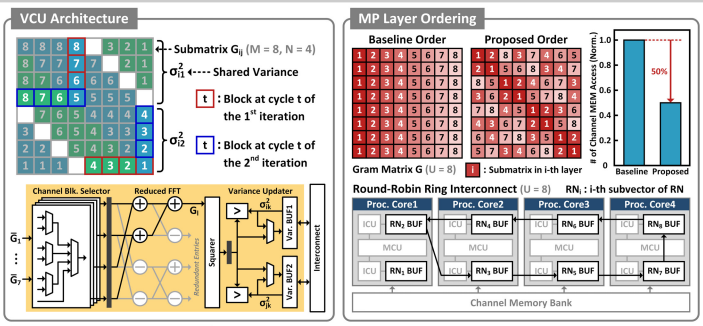
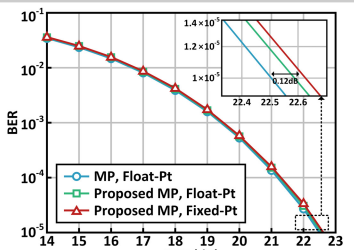



**Figure 2.6.5: Detailed VCU architecture, MP layer ordering, and channel coefficient encoding.**

**System Settings**

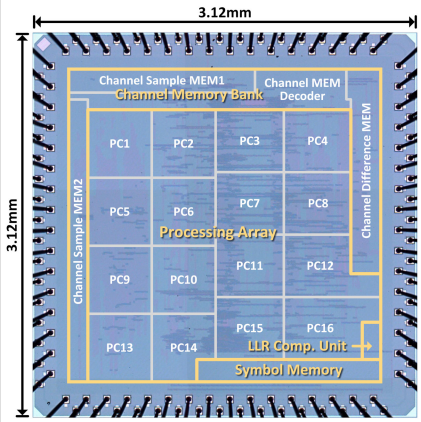
| | |
|---|---|
| MU-MIMO (B × U) | 256 × 32 |
| Modulation | 256-QAM |
| Number of Subcarriers M | 16 |
| Number of OTFS Symbols N | 4 |
| Carrier Frequency | 4 GHz |
| Subcarrier Spacing | 15 kHz |
| Maximal User Speed | 500 km/hr |
| Number of Channel Taps P | 4 |




| | ISSCC'18 [4] | SSCL'19 [5] | VLSI'20 [6] | This Work |
|---|---|---|---|---|
| Technology (nm) | 28 | 28 | 40 | 40 |
| MU-MIMO (B × U) | 128 × 16 | 256 × 32 | 256 × 32 | 256 × 32 |
| Modulation (QAM) | 4-256 | 256 | 4-256 | 256 |
| Supported System | OFDM | OFDM | OFDM | OTFS |
| Soft Output | No | Yes | Yes | Yes |
| Core Area (mm²) | 2.0 | 0.37 | 0.73 | 6.47 |
| Supply Voltage (V) | 1.0 | 0.9 | 1.1 | 0.9 |
| Clock Frequency (MHz) | 569 | 400 | 290 | 200 |
| Power (mW) | 127 | 151 | 87 | 131 |
| Max. Throughput (Gbps) | 1.8 | 0.35 | 1.96 | 6.4 |
| Norm. Energy (pJ/bit)* | 403 | 616 | 44.4 | 20.5 |

* Normalized to 40nm technology node, normalized by $(U/32)^2$

**Figure 2.6.6: Performance evaluation and comparison with the state-of-the-art OFDM detectors.**




| | | |
|---|---|---|
| Technology | | 40-nm CMOS |
| MU-MIMO (B × U) | | 256 × 32 |
| Modulation | | 256-QAM |
| Area ($mm^2$) | Die | 9.73 |
| | Core | 6.47 |
| Gate Count (M) | | 6.76 |
| On-Chip SRAM (KB) | | 167 |
| Supply Voltage (V) | | 0.9 |
| Clock Frequency (MHz) | | 200 |
| Power (mW) | | 131 |
| Max. Throughput (Gbps) | | 6.4 |

**Figure 2.6.7: Chip micrograph and summary.**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## 2.7 BayesBB: A 9.6Gbps 1.61ms Configurable All-Message-Passing Baseband-Accelerator for B5G/6G Cell-Free Massive-MIMO in 40nm CMOS

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The global success of 5G wireless exerts a significant impact on human life [1]. For the forthcoming beyond-5G (B5G) and 6G eras [2], there is demand for wireless networks to evolve from consumer applications towards production-centric demands, to provide comprehensive support to diverse vertical industries. Hence, cell-free massive MIMO [3] has emerged as a crucial catalyst for B5G/6G, delivering magnitudes' higher spectrum efficiency (SE). B5G/6G expects baseband (BB) chips that can simultaneously offer improved throughput (higher than 8Gbps/user), ultra-low latency (less than 2ms), and flexible multi-application support (with exceptional configurability). Existing literature mainly emphasizes individual BB modules [4-6], like MIMO detectors, channel decoders or their combination. Thus, a system-level BB chip that can deliver satisfactory performance in terms of high throughput, low latency, and adaptability to various applications is desirable for B5G/6G [2]. As shown in Fig. 2.7.1, the utilization of such a system-level BB chip poses three challenges: 1) Achieving high throughput is demanding, as it involves designing a high-speed interface that can not only handle the anticipated increase in B5G/6G data traffic, but also adhere to the IEEE 802.3 standard. 2) Ensuring low latency is challenging, because it requires incorporating various algorithms into a single BB chip, while meeting the stringent time requirements of B5G/6G applications. 3) Facilitating exceptional configurability is difficult, as it necessitates a flexible architecture that can serve multiple applications in the B5G/6G realm.

This work presents a 19.093mm$^2$ BB accelerator IC, which delivers a throughput of 9.6Gbps and a latency of 1.61ms. In contrast to reported BB ICs [4-6], the chip integrates a comprehensive BB system encompassing key components, such as a high-speed interface, a channel estimator, a MIMO detector, and a versatile channel decoder, alongside other essential modules. In designing this chip, named BayesBB, we have adopted a message-passing approach based on Bayesian inference, to implement all the BB modules. BayesBB leverages several advantages, including jointly optimized performance, high parallelism, efficient inter-module communication and excellent configurability. BayesBB embodies three features: 1) The high-speed design seamlessly integrates a 10GB Ethernet (XGE) interface with 10.3125Gbps SerDes and 200MHz-matched architectures of a preprocessor together with a channel estimator. This integration enables a throughput of 9.6Gbps, which is 7.1× better than the state-of-the-art [6]. 2) Another aspect of BayesBB is the implementation of an 8×8 16-QAM MIMO belief propagation (BP) detector using a fully unfolded architecture with 5 iterations. This design delivers a latency of 1.61ms, resulting in a 2.5× reduction compared to the user-plane latency (4ms) required by 5G [1]. 3) The configurability of BayesBB allows it to handle various applications of B5G/6G (e.g., decoding both 3GPP LDPC and polar codes). Furthermore, BayesBB's scalability enables it to operate as a 128×128 cell-free massive-MIMO system with commercial B5G/6G configurations (keeping in mind that each BayesBB supports an 8×8 MIMO), achieving more than 100b/s/Hz SE, which is 6.7× higher than the 15b/s/Hz SE required in 5G [1].

Figure 2.7.2 shows the overall architecture of BayesBB, mainly involving Gaussian approximation message passing (GAMP) channel estimation, BP MIMO detection, and BP channel decoding. The chip employs the XGE interface, involving the SerDes, physical coding sublayer (PCS), and media access control (MAC). The XGE interface supports the data transceive with 10.3125Gbps throughput, following the IEEE 802.3 standard. After channel estimation, the MIMO detector recovers the transmitted symbols and sends the bit log-likelihood ratios (LLRs) to the channel decoder, which is configurable to perform both the 3GPP LDPC and polar decoding. The all-message-passing BB algorithm yields up to 2.4dB simulated gains compared to conventional BB algorithms. The measured eye diagram results underscore the high-quality performance of the 10.3125Gbps signal.

Figure 2.7.3 details the high-speed design involving the Ethernet frame parsing, preprocessing, and channel estimation. After parsing the Ethernet frame, the preprocessor buffers the incoming data in the asynchronous FIFO and performs phase compensation together with data descrambling. The channel estimator utilizes the descrambled data to conduct the channel matrix, which will be stored in the output buffer, ready for MIMO detection. To achieve a high-speed design, both the preprocessor and the channel estimator are pipelined architectures. This design achieves a system throughput of 9.6Gbps, which is 7.1× of the prior art [6].

Figure 2.7.4 illustrates the implementation details of the MIMO-BP detector, utilizing a fully unfolded architecture with 5 iterations. The Gaussian approximation of interference is first measured to support the posterior message calculation. Then, the posterior messages are buffered and accumulated to compute the symbol prior probability. Finally, bit LLR messages are derived from the symbol prior probability. The detector benefits from a fully unfolded and pipelined architecture, enjoying a detection latency of 1.44ms. The low-latency detector results in a system latency of 1.61ms, which represents a 2.5× reduction compared to the user latency of 5G [1].

Figure 2.7.5 shows the configurable architecture of the channel decoder. The 32 sub-band data processed by the MIMO detector are transmitted to the decoder and buffered in the storage array after data-dimension conversion and rate matching. BayesBB incorporates two decoder modes, LDPC and polar decoding. The decoding mode, iteration counts, CRC modes, and other state messages can be managed through the configurable register using the SPI serial bus interface. The LDPC decoder adopts a block-based and row-layered update architecture, with block parallelism of 80 for the code length of 1760b to support the Base Graph 2 in 5G NR. To ensure system bandwidth, two LDPC decoders are instantiated to correspond to the divided sub-band data. The polar decoder employs a single-column architecture with a code length of 2048b in 5G NR. After the final decoded result is obtained, it will be verified by the CRC module.

Figure 2.7.6 summarizes the highlights of BayesBB (left) and demonstrates its merits compared with prior art [4-6] (right). BayesBB is an all-message-passing full-BB-processing chip for B5G/6G commercial system. It configures decoding modes, error-checking and interface settings for high-speed transmission in various scenarios, such as 4K ultra-HD video streaming, unmanned aerial vehicle (UAV) applications and autonomous driving. The scaled-up 16 BayesBB array can also accommodate up to 128×128 antennas with 3,200 subcarriers for 153.6Gbps throughput in actual system testing (left). Thanks to the soft output interface between the detector and the decoders, a single BayesBB achieves 7.1× higher throughput and 2dB bit error rate (BER) gains over [6]. The peak throughput reaches 9.6Gbps at 200MHz system frequency. The detector achieves 24.8× throughput, 2.1× area efficiency and 4.6× energy efficiency compared to [5] (upper right). The decoders demonstrate comprehensive superiority in both LDPC and polar decoding modes (bottom right). The LDPC decoder boosts throughput by 3.2× and reduces power by 41.1% and area by 34.1% area vs. [6]. The polar decoder improves both area and energy efficiency by an order of magnitude compared to previous work [4-6]. In LDPC and polar decoding modes, the decoders outperform the prior art [6] in terms of area efficiency by 9.4× and 3.4×, and energy efficiency by 10.3× and 7.8×, respectively.

Figure 2.7.7 presents the die micrograph and its specification summary. The chip fabricated in 40nm CMOS occupies a die size of 19.093mm$^2$ with 3.2W. The power consumption can be reduced by multi-voltage domain technology with 1.1V for the digital, 1.1V and 2.5V for SerDes PMA and 3.3V for IO supply.

*Acknowledgement*:
This work was supported in part by NSFC (62122020, 62331009, and 62225107), in part by National Key R&D Program of China (2020YFB2205503), in part by NSF of Jiangsu Province (BK20211512), in part by Major Key Project of PCL (PCL2021A01-2), and in part by Fundamental Research Funds for the Central Universities. Corresponding authors: Xiaohu You and Chuan Zhang ({xhyu, chzhang}@seu.edu.cn).

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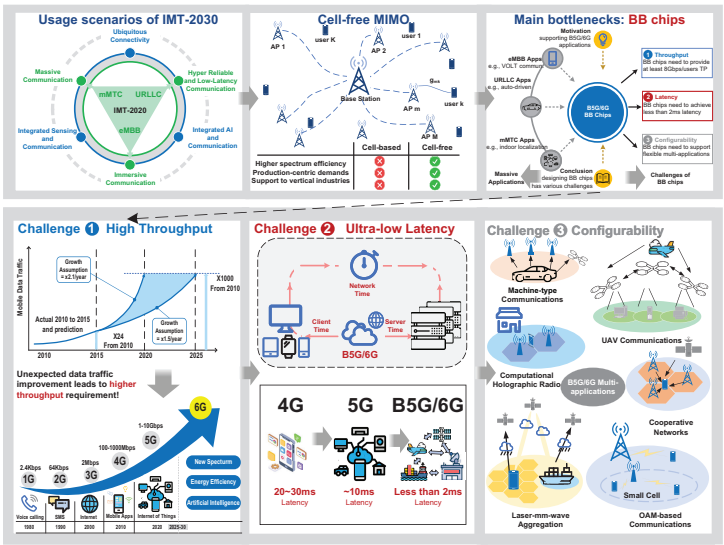



Figure 2.7.1: Designing a system-level baseband chip is challenging.

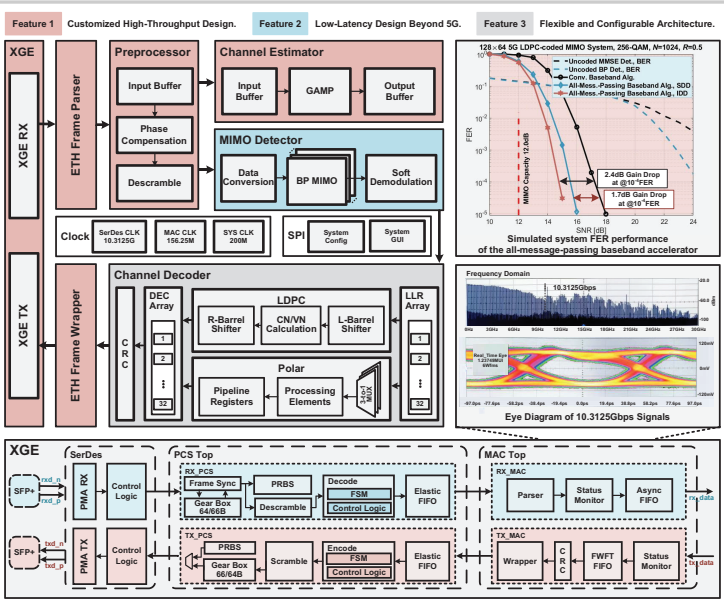



Figure 2.7.2: Overall architecture of BayesBB.

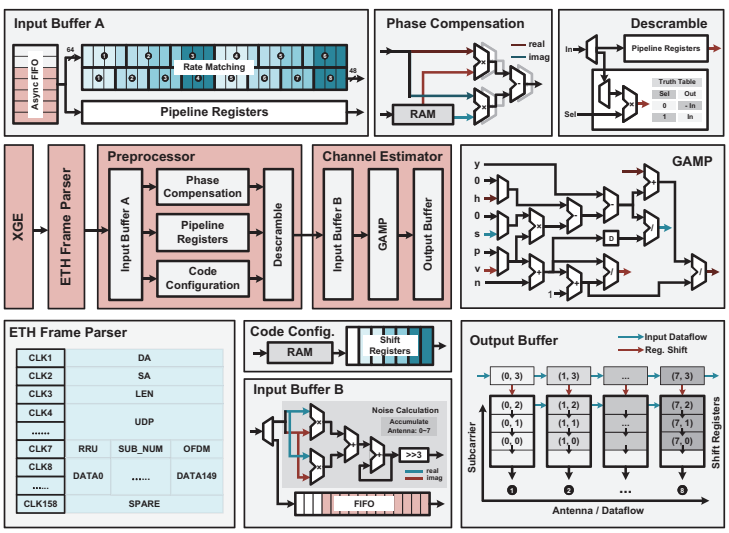



Figure 2.7.3: High-throughput design for preprocessor and channel estimator.

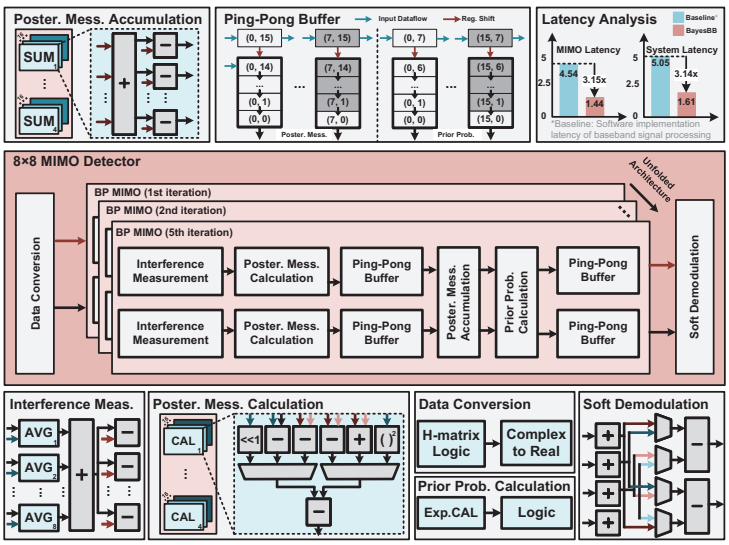



Figure 2.7.4: Low-latency fully unfolded architecture of MIMO detector.

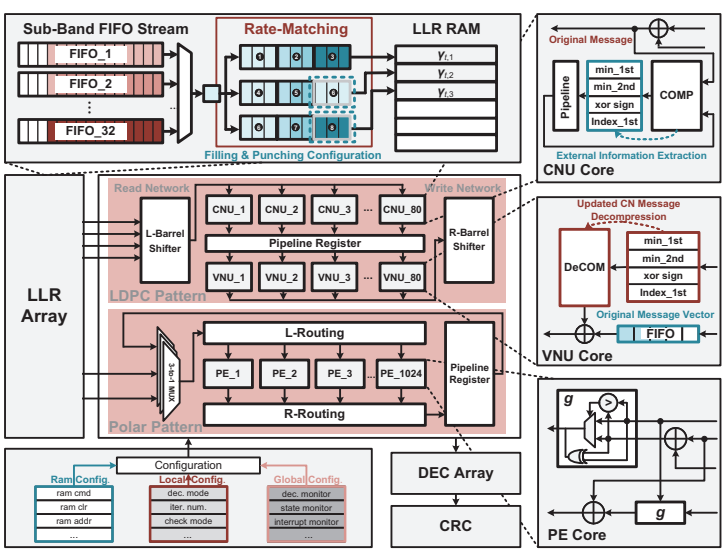



Figure 2.7.5: Configurable architecture of channel decoder.

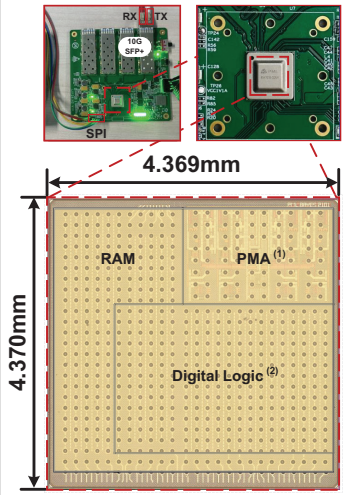
| | |
|---|---|
| System | Commercial B5G/6G cell-free massive MIMO system |
| Core Functions | ✓ **High-speed interface function:** Ten-Gigabit Ethernet (XGE) with IEEE 802.3 standard; 5G/NR protocol stack<br>✓ **Data processing:** Data analysis; Wrapping; Descrambling; Phase compensation<br>✓ **Baseband processing:** Channel estimation; MIMO detection; LDPC/polar decoding; CRC check; SPI bus configuration; Ping-pong buffer |
| Configurability | ✓ **All the following can be configured:** Data processing for different users; Decoding modes; Iteration counts; CRC check modes; Descrambling and phase compensation parameters; Signal quality adjustment parameters; Frame setting parameters; Multi-loopback modes; Status monitoring modes |
| Scalability | Single chip supports 8x8 antenna, 800 subcarriers; Support multiple-chip collaboration; Testing with a maximum of 128x128 antenna, 3,200 subcarriers for a throughput of 153.6Gbps |
| Applications | 4K ultra HD video streaming; Unmanned aerial vehicle (UAV) applications; Autonomous driving; Virtual & augmented reality |
| System Interfaces | Optical port; XGE including SerDes, PCS, and MAC; SPI serial bus |
| New Concept | ✓ All-message-passing BB signal processing |

| Detector | This Work | ISSCC'12 [4] | ISSCC'14 [5] | JSSC'19 [6] |
|---|---|---|---|---|
| Algorithm | BP (1) | SD | SD | MMSE |
| Output Interface | Soft | Hard | Hard | Hard |
| MIMO System | 8x8 (2) | ≤4x4 | ≤4x4 | 4x4 |
| Modulation | 16-QAM | ≤64-QAM | ≤64-QAM | 256-QAM |
| Technology (nm) | 40 | 65 | 65 | 65 |
| Core Area (mm²) | 4.36 | 0.31 | - | 0.7 |
| Frequency (MHz) | 200 | 333 | 445 | 517 |
| Power (mW) | 456 | 38 | 87 | 26.5 |
| SNR @ BER=10⁻³ (dB) (3) | 13 | 12.5 | 12.5 | 15 |
| Throughput (Mb/s) | 9,830 | 296~807 | 396 | 1,379 |
| Area Efficiency (Mb/s/kGE) | 2.16 | 1.37~3.75 | 1.03 | 3.68 |
| Energy Efficiency (pJ/b) | 47.5 | 48 | 220 | 19.2 |

| Decoder | This Work | | ISSCC'12 [4] | ISSCC'14 [5] | JSSC'19 [6] |
|---|---|---|---|---|---|
| Codes | LDPC | Polar (4) | LDPC | LDPC | NB-LDPC |
| Block Length | 3,520 (5) | 2,048 | 768 | 768 | 416 |
| Technology (nm) | 40 | | 65 | 65 | 65 |
| Core Area (mm²) | 0.58 | 1.62 (6) | 3.6 | - | 1.7 |
| Frequency (MHz) | 200 | 200 | 267 | 500 | 307 |
| Power (mW) | 60.7 | 170 | 367 | - | 103 |
| Iterations | 10 (7) | 20 (7) | 10 | 10 | 5-10 |
| Throughput (Mb/s) | 3,276.8 | 3,276.8 | 235.2 | 155 | 512~1,024 |
| Area Efficiency (Mb/s/mm²) | 5,649.66 | 2,022.72 | 65.33 | 100.9 | 301~602 |
| Energy Efficiency (pJ/b/iter) | 1.85 | 2.59 | 170 | 232 | 20.1 |

(1) Message-passing detection. (2) Scalable design, supporting a maximum of 128x128 MIMO in testing. (3) Scaled to 8x8 16-QAM MIMO scenario. (4) The configurable decoding mode for LDPC codes and polar codes. (5) Composed of two LDPC decoders with the code length 1,760. (6) CRC module is included for early check. (7) The number of iterations can be configured.

Figure 2.7.6: Main highlights and performance comparisons with prior art.




| | Specifications |
|---|---|
| Technology | 40nm CMOS |
| Application | Baseband Processing |
| Die Size (mm²) | 19.093 |
| Core Power (W) | 3.2 |
| Package | Flip Chip BGA |
| Serial | SPI |
| RAM Size(Mb) | 4.1 |
| Logic Gate Count | 15,572,564 |
| Digital Logic Voltage (V) | 1.1 |
| SerDes PMA Voltage (V) | 1.1 and 2.5 |
| IO Voltage (V) | 3.3 |
| System Frequency (MHz) | 200 |
| MAC Side Frequency (MHz) | 156.25 |
| Max. System Throughput (Gbps) | 9.6 |

(1) Physical Medium Attachment (PMA) is the analog core in SerDes.
(2) Digital logic of the BayesBB.

**Figure 2.7.7: Die micrograph and design summary.**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## 2.8 A 21.9ns 15.7Gbps/mm² (128, 15) BOSS FEC Decoder for 5G/6G URLLC Applications

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To enable emerging mission-critical applications, e.g., healthcare monitoring, remote surgery, and autonomous driving, 5G/6G ultra-reliable low-latency communication (URLLC) devices demand the concurrent fulfillment of ultra-reliability, low-latency, and low-power communications, particularly in short data transmissions as depicted in Fig. 2.8.1. However, the existing short-length forward error-correction (FEC) solutions for URLLC devices cannot meet all the challenging requirements at the same time. The recent short block-length and low-rate block orthogonal sparse superposition (BOSS) code [1] has been proposed as one of the promising next-generation FEC candidates to support all the URLLC features with its low latency and strong reliability. This work presents an integrated BOSS decoder as an URLLC FEC solution by fully capitalizing on the BOSS code's robust error correction and low latency. Our design achieves enhanced area and energy efficiencies at a low area cost, outperforming the state-of-the-art URLLC FEC designs [2-6].

Figure 2.8.1 illustrates how to encode and decode 15 message bits and 3b cyclic redundancy check (CRC) in a (128, 15) BOSS codeword [1]. Utilizing 32 128×128 unitary matrices as a codebook, the BOSS encoding selects one matrix and its two columns by using the given 18 input bits as selection indices. The final encoded 128×1 codeword is calculated by a superposition of two selected columns. For the received codeword from the noisy channel, the BOSS decoder estimates the selection indices of the matrix and columns to recover the 15 message bits. In contrast to previous URLLC FEC decoders based on sequential or iterative algorithms [2-6], the proposed BOSS decoder is feedforward and highly parallelized, as shown in Fig. 2.8.1, leading to a significantly lower latency, e.g., 6.2× and 3.1× lower latency than the 5G polar [2] and LDPC [3] designs, respectively. A parallel BOSS decoder, however, comes with a high area cost to compute 32 128×128 by 128×1 matrix-vector multiplications (MVMs) and 32 128-input sorting operations required for the (128, 15) BOSS code. In this work, we present two cost-efficient circuits to make our parallel BOSS decoder practical: 1) Fast Walsh-Hadamard transform (FWHT)-based MVM calculator and 2) iterative max-min tree (IMMT), which reduce the decoder area by 86% and then an additional 68%, respectively, resulting in the superior energy and area efficiencies for URLLC scenarios.

Figure 2.8.2 shows our 4× parallel BOSS decoder architecture that is time-multiplexed by 8× to complete the decoding as shown in its pipelined processing flow. The architecture consists of 4 lanes divided into 3 pipeline stages: 1) FWHT-based MVM stage with 4 parallel computing lanes, 2) maximum a-posteriori (MAP) decoding stage with 4 IMMTs per lane, and 3) output generation stage that utilizes a score calculation unit (SCU) per lane to identify the most reliable candidate. In the first stage, more specifically, each lane activates a permuting unit (PU) (among the dedicated 8 PUs) and the FWHT unit per cycle, which computes a FWHT-based MVM. Over 8 clock cycles, the 4 lanes perform 32 MVMs between one 128×1 received vector ($\mathbf{r}$) and 32 128×128 unitary matrices ($\mathbf{U}_0$-$\mathbf{U}_{31}$) to produce 32 128×1 products ($\mathbf{U}_x\mathbf{r}$). In the second MAP stage, taking 4 products from the prior stage per cycle, 4 IMMTs are activated at a time, as depicted in the flow chart, to generate 16 message candidates (4 per product). A total of 128 candidates are generated throughout the second stage. In the last stage, 4 SCUs receive 16 candidates in every cycle from the IMMTs and calculate their reliability scores with CRC validation. The scores are sent to the maximum comparator to find the most reliable 15b message output among the 128 candidates. Using the balanced pipelined processing, as shown in Fig. 2.8.2, it takes only 13 cycles to decode a BOSS codeword. The decoder offers robust error correction by checking up to 128 possible output candidates.

Figure 2.8.3 illustrates the proposed FWHT-based MVM calculator. To reduce the hardware complexity of compute-intensive general MVM calculators, which are required by the original BOSS decoding [1], we design the low-cost FWHT-based MVM calculators by using three optimization methods. The first method leverages the property that the permutation of a unitary matrix is also unitary; and therefore, we construct 32 different unitary matrices ($\mathbf{U}_0$-$\mathbf{U}_{31}$) by multiplying one base matrix ($\mathbf{H}$) by 32 different permuting matrices ($\mathbf{P}_0$-$\mathbf{P}_{31}$). To construct a parallel decoder, we instantiate $N_F$ (≤32) MVM calculators shared by $32/N_F$ PUs as shown in Fig. 2.8.3, where $N_F$ is a parallelization factor. Then, the permutation-based unitary matrices allow replacement of general MVM calculators with constant MVM calculators (with the fixed matrix input $\mathbf{H}$), saving area by 51%. To further reduce the computational complexity, in addition, we introduce two constraints for designing the cost-efficient base matrix $\mathbf{H}$: making it: 1) a Hadamard matrix and 2) block-diagonal. Selecting $\mathbf{H}$ to be a 128×128 Hadamard matrix permits the use of cost-efficient FWHT processing to perform the proposed constant MVM. Due to unique properties of the Hadamard matrix, the proposed FWHT-based MVM calculator is realized by only simple adders without using multipliers, reducing the hardware cost by 95%. By introducing the block-diagonal unitary matrices shown in Fig. 2.8.3, we can decompose a 128×128 FWHT operation into 16×16 ones, further reducing area by 43%. Using low-cost MVM calculators, we finally investigate the tradeoff between the overall decoder latency and the area by varying $N_F$. Selecting $N_F$=4 minimizes the area-time (AT) complexity of the prototype decoder, saving overall area by 86% compared with a baseline architecture using general MVM calculators.

Figure 2.8.4 illustrates the 4-phase IMMT circuit for the second decoding stage. The goal is to search for 4 max-min pairs per $\mathbf{U}_x\mathbf{r}$, i.e., ($I_{max1}$, $I_{min1}$), ($I_{max2}$, $I_{min2}$), ($I_{max3}$, $I_{min3}$), and ($I_{max4}$, $I_{min4}$) as the message candidates, where $I_{maxk}$ (or $I_{mink}$) denotes the index of maximum (or minimum) element in $\mathbf{U}_x\mathbf{r}$ to generate the $k$th pair. According to [1], $I_{maxk}$ and $I_{mink}$ cannot be all even or odd values at the same time. The naïve architecture for the searching task is to use compute-intensive bitonic sorters, and after adopting the FWHT-based calculators, the sorters dominate the overall area cost. To address the huge cost problem, IMMTs adopt an iterative searching architecture that generates 4 max-min pairs sequentially in 4 cycles. As depicted in Fig. 2.8.4, an IMMT consists of an index checker, 64 compare-and-swap (CAS) units, a 64-to-1 max-tree, and a 64-to-1 min-tree. The index checker removes the prior max/min values, so that the following IMMT phases can find the next max-min pairs properly. The 64 CAS units make two 64-size outputs from the 128×1 product result of $\mathbf{U}_x\mathbf{r}$, which are fed to searching trees to find a max, a min, and their indices. To get valid max-min pairs, the IMMT offers two computing modes by dynamically changing the inputs of the min-tree according to the previous results. As shown in Fig. 2.8.4, the mode-1 processing is defined for the first IMMT phase, activating the min-tree with 64 values transmitted from CAS units. The mode-2 processing is defined for the remaining IMMT phases, simply passing odd-indexed (or even-indexed) values to the min-tree depending on the prior max indices. The two computing modes make it possible to find 4 valid pairs in 4 cycles without additional cycles. Choosing the number of IMMTs ($N_T$) is important for achieving the best tradeoff point between latency and complexity. As shown in Fig. 2.8.4, the proposed decoder uses 16 IMMTs, which reduces the overall area by 68%, while maintaining the same latency as the baseline design that uses bitonic sorters.

Figure 2.8.5 shows the measurement results of the prototype BOSS decoder fabricated in 28nm CMOS. At the nominal supply voltage of 0.95V, the decoder achieves the low latency of 21.9ns to decode a (128, 15) codeword consuming 33.3mW, where the latency is shortened to 13.7ns at the maximum voltage of 1.15V. In terms of frame error rates (FERs), even with the various cost-optimizations, the BOSS decoder successfully offers similar correcting performance as the baseline algorithm [1]. Note that this work outperforms existing short-length FEC solutions [2-3], providing a comparable FER to the medium-length LDPC code [4]. Figure 2.8.6 details the implementation results of recent URLLC FEC decoders [2-6]. For a fair comparison, we additionally estimated the results of [2] at the Eb/N0 of 5dB with the reported algorithm and design. In this case, its latency exponentially increases due to the large number of test error patterns at low Eb/N0. This work requires Eb/N0 of 5.2dB for the target FER of $10^{-5}$ and achieves low latency, area-efficiency of 15.78Gbps/mm², and energy efficiency of 5.7pJ/b compared to [2-6]. As shown in Fig. 2.8.5, this work achieves 6.2× lower latency and 2× better energy-efficiency than a recent polar decoder [5] for the same code configuration. Note that this work offers deterministic latency regardless of the channel condition, which is 1.95× faster than the GRAND decoder under high Eb/N0 (7dB) [2]. Figure 2.8.7 shows the chip micrograph and area/power breakdowns of the BOSS decoder, which occupies a small core area of 0.37mm².

*Acknowledgement:*
This work was supported by the National Research Foundation funded by the Korea government (2022R1A2C2092521), the Grand Information Technology Research Center support program (IITP-2023-2020-0-01612), Samsung Electronics Co., Ltd., and IDEC.

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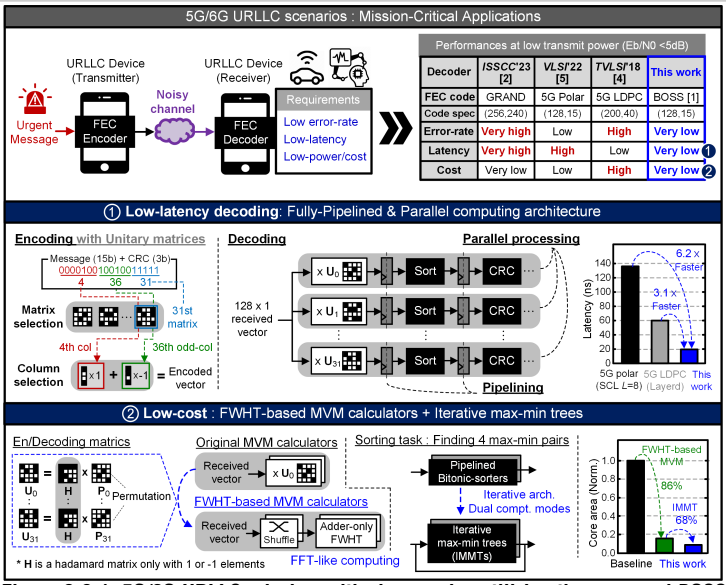


Figure 2.8.1: 5G/6G URLLC mission-critical scenarios utilizing the proposed BOSS FEC decoder.

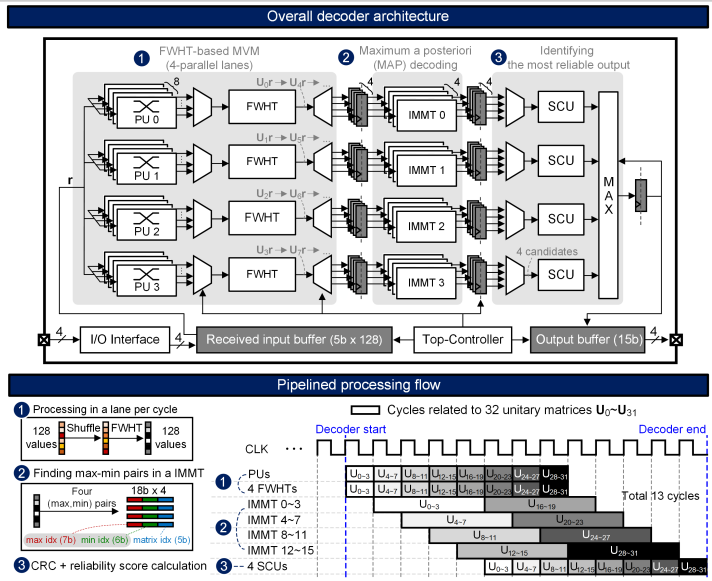


Figure 2.8.2: Overall decoder architecture with 4 parallel computing lanes and its processing flow.

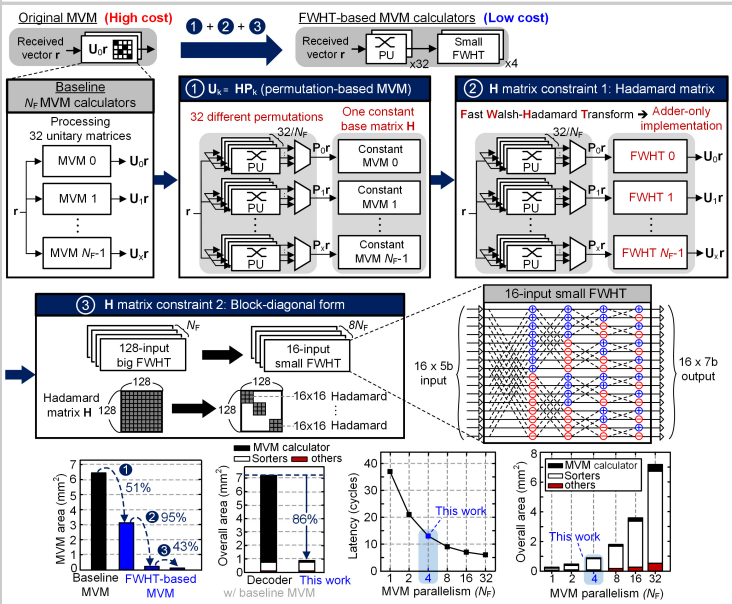


Figure 2.8.3: Implementation details of the FWHT-based MVM calculators.

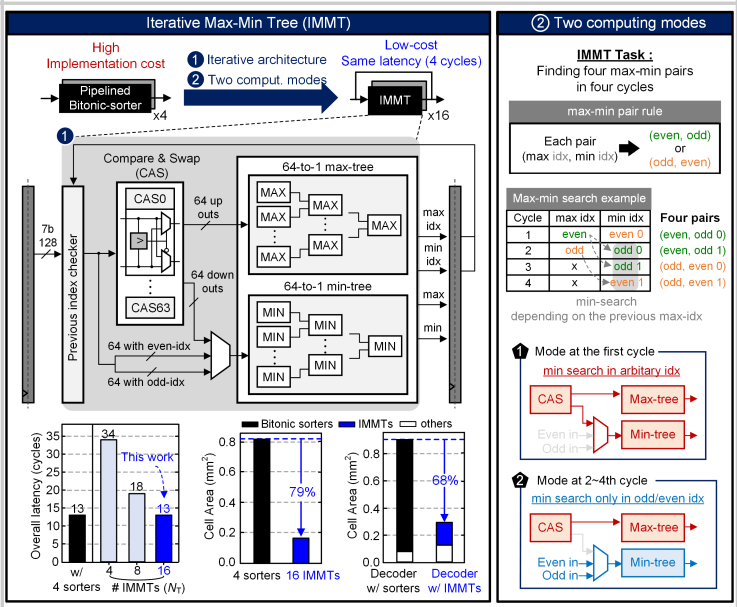


Figure 2.8.4: Implementation details of the iterative max-min trees (IMMTs).

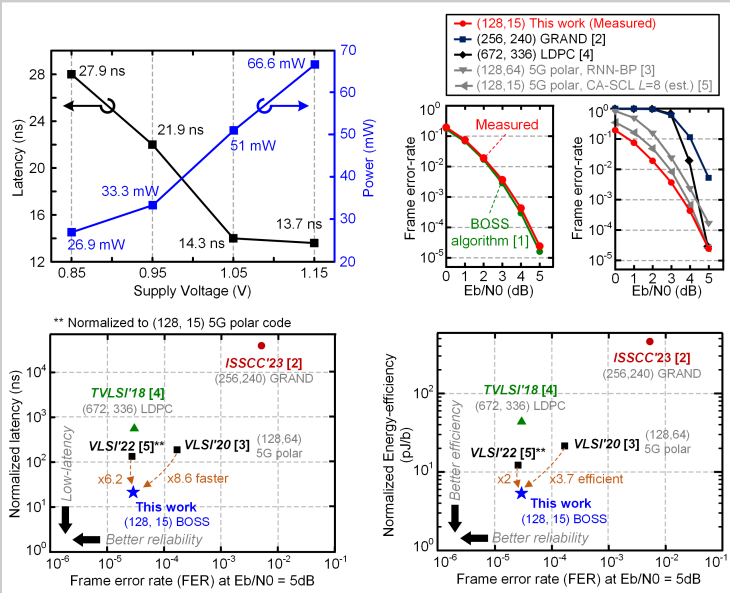


Figure 2.8.5: Measurement results.

| | This work | ISSCC'23 [2] | VLSI'20 [3] | | | TVLSI'18 [4] | VLSI'22 [5] | JSSC'21 [6] |
|---|---|---|---|---|---|---|---|---|
| Technology | 28nm | 40nm | 40nm | | | 28nm | 28nm | 40nm |
| FEC code | BOSS code | GRAND + Polar | Polar code | | | LDPC | Polar code | Polar code |
| Decoding Algorithm | BOSS dec. | ORBDRAND | RNN-BP | | | MS | CA-SCL | CA-SCL |
| Code length | 128 | 256 | 256 | 128 | 64 | 672 | 1024 | 1024 |
| **Eb/N0 @ FER=$10^{-5}$** | **5.2** | 7 | 5.2 | 5.8 | 6.3 | 5.2 | 2.75 | 3.6 |
| Core area ($mm^2$) | 0.37 | 0.4 | 0.18 | | | 1.99 | 0.595 | 0.317 |
| Supply Voltage (V) | 0.95 | 1.0 | 0.9 | | | 0.9 | 1.05 | 0.9 |
| Frequency (MHz) | 590 | 90 | 225 | | | 150 | 413 | 476 |
| Power (mW) | 33.3 | 4.8 | 12.8 | | | 399 | 101.4 | 42.8 |
| **Constant latency** | **O** | X | O | | | X | O | O |
| **Latency (ns)** † | **21.9** | 59986* | 61.3** | 310 | 270 | 220 | 555 | 1100 | 2260 |
| Throughput† (Gbps) | 5.84 | 0.004* | 4.3** | 0.82 | 0.47 | 0.29 | 9.69 | 0.93 | 0.45 |
| **Area-efficiency† ($Gbps/mm^2$)** | **15.78** | 0.011* | 10.8** | 4.57 | 5.33 | 6.4 | 3.41 | 1.56 | 4.17 |
| **Energy-efficiency† (pJ/b)** | **5.7** | 1125* | 1.14** | 13.6 | 11.7 | 9.7 | 41.2 | 109.5 | 89.9 |
| **Normalized to 28nm, 0.95V** ‡ | | | | | | | | | |
| Latency (ns) | 21.9 | 41990* | 42.9** | 217 | 189 | 154 | 555 | 1100 | 1582 |
| Throughput (Gbps) | 5.84 | 0.006* | 6.14** | 1.17 | 0.67 | 0.41 | 9.69 | 0.93 | 0.65 |
| Area-efficiency (Gbps/mm^2) | 15.78 | 0.031* | 31.3** | 13.3 | 7.6 | 4.7 | 3.41 | 1.56 | 2.22 |
| Energy-efficiency (pJ/b) | 5.7 | 497.4* | 0.50** | 12.1 | 21.1 | 34.3 | 45.90 | 89.64 | 49.08 |

* is estimated at Eb/N0 = 5 dB
** is measured at Eb/N0 = 7 dB
† is measured for decoding a frame
‡ Norm. Latency= Latency * X
Norm. Area-effi. = Area-effi. / $X^3$
Norm. Throughput = Throughput / X
Norm. Energy-effi.= Energy-effi. * $X^2Y^2$
where X = 28nm / used tech. , Y = 0.95V / used voltage

Figure 2.8.6: Comparison table.

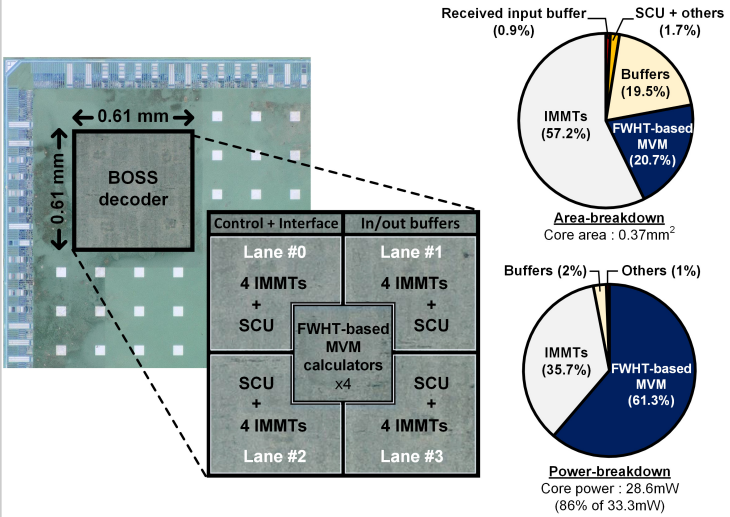








**Figure 2.8.7: Chip micrograph and area/power breakdowns.**


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